Arrears as Borrowing: The Economics of Partial Default*

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Abstract

Sovereign defaults are rarely complete: governments withhold part of their debt service, the missed payments accrue as arrears, and countries often leave a default episode owing more than when they entered it. Arellano, Mateos-Planas, and Ríos-Rull (2023) built the theory of *partial default* around these facts. We take their model as the right description of sovereign distress and study it in continuous time, where it becomes tractable enough to say precisely what partial default does. Three things stand out. Arrears are borrowing: withholding debt service issues new debt to existing creditors at an implicit price that the market does not set, paid for with the output cost of default. The default decision is therefore a static financing decision, a sovereign never defaults a little, and in bad times it targets more debt, not less; after a negative shock debt rises for years through arrears while the sovereign stays in the market. Second, the model has a genuine pathology at high debt: where default is cheap, a sovereign in full default is financed without limit, and what looks like a second steady state in computed solutions is the entry point of an explosion. Third, an underwriting standard on lenders is the minimal repair: it makes the equilibrium well posed, and the sovereign defaults more where it binds. The model fits the data targets of the discrete-time calibration well and reproduces the protracted, debt-increasing default episodes, with closed-form default thresholds and debt limit, exact long-run distributions and impulse responses, and no need for auxiliary smoothing shocks.

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1 Introduction

Sovereign defaults are rarely complete. Governments in distress typically continue to service some of their obligations while suspending payments on others, missed payments accumulate as arrears that add to the stock of debt, and countries frequently emerge from a default episode owing more than when it began. Arellano, Mateos-Planas, and Ríos-Rull (2023) document these facts and build a theory of *partial default* around them: a sovereign that chooses, period by period, what share of its debt service to withhold, at a cost that rises with the share withheld. Their framework replaces the binary default decision of the standard model, introduced by Eaton and Gersovitz (1981) and quantified by Aguiar and Gopinath (2006) and Arellano (2008), with a continuous one, and it replaces exclusion from financial markets with continued access at prices that reflect the expected intensity of default.[1] In their data emerging economies are in partial default about a third of the time and withhold about forty percent of the debt service when they are, and their calibrated model reproduces the frequency, the depth and the dynamics of these episodes. We regard partial default as the right description of sovereign distress and the natural baseline for the quantitative theory of sovereign debt, and this paper is an attempt to push that agenda forward.

We formulate the partial default model in continuous time and study it analytically and quantitatively. The formulation is natural, since the default intensity is a flow decision and arrears accrue continuously, and it is tractable in a way the discrete-time model is not: because the sovereign never loses market access there is no stopping problem, and a single Hamilton–Jacobi–Bellman (HJB) equation for the value function together with a single linear equation for the bond price characterize the equilibrium at every state. That tractability is what lets us say precisely how partial default works. Three things stand out, and they organize the paper.

The first is that *arrears are borrowing*. Write the addition to the stock of debt as issuance plus arrears. Holding that addition fixed, withholding a unit of debt service raises consumption by one minus the market value of the arrears it creates, $1 - \kappa q$: it is the issuance of κ units of debt to existing creditors at an implicit price $1/\kappa$, which exceeds the market price and does not depend on it, paid for with the output cost of default (Proposition 2). This identity has consequences that the discrete-time simulations contain but do not isolate. The default decision is static: given debt, productivity and the price, the share withheld maximizes a concave resource function and does not depend on the value function or on the direction in which debt is moving (Theorem 3), so the regions of no, partial and full default are explicit in the state and the price (Proposition 4). A sovereign never defaults a little: the first default is a jump of known size, half of the debt service at the calibrated parameters, because a small default is not worth its fixed cost. The debt a sovereign is willing to service in full at the risk-free price is a closed-form limit (Theorem 14),

1. Textbook treatments of the standard framework are Uribe and Schmitt-Grohé (2017) and Aguiar and Amador (2021); Aguiar et al. (2016) survey its quantitative extensions. In models with renegotiation, following Yue (2010) and Pitchford and Wright (2012), the sovereign returns to the market with a debt level set by bargaining, as in Mihalache (2020) and Dvorkin et al. (2021).

and the limit is a soft edge: above it the sovereign repays at a speed that vanishes like the cube root of the excess debt while the price leaves par with a vertical tangent (Theorem 16). And because withholding payments is a source of funds, the sovereign targets more debt in bad times, not less; after a negative productivity shock debt rises for years through arrears while the sovereign remains in the market, which is the shape of the episodes in the data (Section 4.3).

The second is that the model has a *genuine pathology at high debt*, which anyone who computes it meets and which the continuous-time formulation lays bare. In states where productivity is low enough that default carries no fixed cost, a sovereign in full default pays nothing, bears only the variable cost, and faces a marginal cost of debt close to zero, while lenders still pay a positive price because productivity will recover. The sovereign then issues at any positive price and debt grows without bound. The entry point of this region is a stationary point of debt that is approached from below and left from above, and the price jumps there: on a grid it looks like a second steady state with a cliff next to it, and the share of the long-run distribution that enters it is decided at the scale of the grid. We show that this is a property of the equilibrium, not of any discretization, that it, and not the knife-edge autarky equilibrium that one might suspect, is what makes computed solutions depend on the upper end of the debt grid, and that no refinement fixes it (Section 3.2).

The third is that an *underwriting standard* on lenders is the natural and minimal repair. Lenders do not absorb new issues at prices below a floor, a restriction of the kind the seniority and restructuring literature has invoked for other purposes (Chatterjee and Eyigungor 2015). The standard makes the full-default region transient, the price continuous and the support of debt bounded independently of the grid (Propositions 11 and 12), and it changes the default decision where it binds in an interpretable way: the sovereign applies the same static rule at its own shadow price of debt instead of the market price, and defaults more (Proposition 10). It binds in deep distress only, on about two percent of the ergodic mass at the calibrated parameters, and it has a reading in terms of market access: primary-market issuance stops at a spread.

The model is also fast and exact to compute, and it matches the discrete-time model quantitatively. Calibrated to the eleven moments that Arellano, Mateos-Planas, and Ríos-Rull (2023) target, with their free parameters and their weights, it fits the data well under their own loss and reproduces the dynamics of partial default episodes: after a negative shock the sovereign withholds a growing share of its debt service, debt rises through arrears for several years, the spread rises, and the episode ends as productivity recovers. The computation is an implicit upwind finite difference scheme in the tradition of Achdou et al. (2022), with the value and the price updated jointly, and the default decision needs no smoothing: the intensity is a continuous control, so the price schedule is continuous wherever the state moves through a point. Discrete-time models with long-term debt rely on i.i.d. shocks to the default and borrowing choices to make the price schedule converge (Chatterjee and Eyigungor 2012; Dvorkin et al. 2021; Mihalache 2020); these devices are widely used and effective, yet they carry both a conceptual cost, the smoothed decision is not the one the model describes, as well as a practical one, convergence that depends on

their scale (see Mihalache (2025) for a survey). In our implementation, the stationary distribution and the impulse responses are computed exactly from the generator of the discretized process rather than by simulation, and the same scheme solves the deterministic, chain, and diffusion cases, which is useful because the first two are where the mechanisms are visible analytically.

The analysis rests on results of independent interest. The static default theorem and the closed-form thresholds hold for any productivity process. The deterministic model has a unique no-default steady state in closed form, approached along a square-root law from below and a cube-root law from above, with a two-term expansion that matches the computed solution; the right branch exists and is unique in a natural class. The bond price is continuous wherever the state moves through a point and can jump only at stationary points that are not approached from both sides, which is the mechanism of the price cliffs; the value is monotone in debt with buy-backs and with the standard. Self-fulfilling partial-default steady states, in which a pessimistic price sustains the default intensity that justifies it, exist as local candidates in a continuum, and whether one is reached is a two-point problem decided by the impatience wedge against the spread the first default implies; at the calibrated parameters none is, and we record the general case as a conjecture.

Our paper belongs to the recent literature on quantitative sovereign default in continuous time. Bornstein (2020) compares discrete and continuous time versions of the standard model, finds that computation is faster in continuous time, and identifies a *painful deleveraging* effect specific to settings in which the stock of debt is slow-moving.[2] Tourre (2017) studies the macro-financial implications of the standard model with recursive utility and a rich pricing kernel. Hurtado, Nuño, and Thomas (2023) is the paper closest to ours: they study a sovereign with long-term debt denominated in local currency, who can reduce the real value of the debt through inflation. Inflation lowers the real value of the entire stock of debt, whereas partial default applies to the debt service flow and raises the stock through arrears. All three papers solve optimal stopping problems with HJB variational inequalities, because default is a discrete transition into exclusion; the partial default model has no such transition, which is the source of its tractability.

Continuous-time methods have also been at the forefront of theoretical work on sovereign debt. Aguiar and Amador (2020) study multiplicity and debt maturity in a continuous-time model with a binary default decision, and Amador and Phelan (2021, 2023) study reputation, the latter in a setting with partial default. The slow-moving debt crises of Lorenzoni and Werning (2019) are the closest relatives of the self-fulfilling partial-default steady states we discuss in Section 3.3: in our model a pessimistic price sustains a higher default intensity which sustains the price, and whether such a steady state can be reached from low debt depends on the sovereign's willingness to borrow into default at a discount. The underwriting standard we adopt is a restriction on lenders of the kind studied by Chatterjee and Eyigungor (2015), who show that a seniority arrangement that limits what lenders absorb can improve on unrestricted markets; in

2. Rendahl (2022) discusses the extent to which similar results can be obtained in discrete time by adapting the sparse matrix methods at the core of the continuous-time solution.

our setting the standard has the additional role of making the equilibrium well posed at high debt.

On the computational side we follow Achdou et al. (2022) and Fernández-Villaverde, Hurtado, and Nuño (2023). The partial default model requires two additions to their scheme: a joint implicit update of the value and the price, because the alternating iteration between them is unstable, and a monotone treatment of the non-concave Hamiltonian created by the fixed cost of default. We also report, in Appendix D, an attempt to solve the diffusion case with physics-informed neural networks (Raissi, Perdikaris, and Karniadakis 2019; Sirignano and Spiliopoulos 2018), the continuous-time counterpart of the deep learning methods of Maliar, Maliar, and Winant (2021) and Azinovic, Gaegauf, and Scheidegger (2022). The attempt fails for an instructive reason: the stationary residual that these methods minimize does not identify the economically relevant solution, and the networks converge to spurious solutions with smaller residuals than the true one.

The rest of the paper proceeds as follows. Section 2 presents the model, derives the HJB and pricing equations, defines the equilibrium and discusses the key assumptions, including the underwriting standard. Section 3 characterizes the equilibrium: the general results, with arrears as borrowing and the static default decision first, then the behavior at high debt, where the underwriting standard enters, then the deterministic and the chain cases, each with a numerical illustration. Section 4 contains the quantitative analysis of the diffusion case: calibration, computation, moments and impulse responses. Section 5 concludes. The appendices contain the proofs, the numerical implementation, robustness checks and the neural network experiment.

2 The Partial Default Model

The economy consists of a small open economy run by a risk-averse sovereign and of a unit measure of competitive, risk-neutral international lenders. Time is continuous. We describe the two problems in turn and define the equilibrium.

2.1 The Sovereign's Problem

The sovereign receives an endowment stream, borrows abroad with long-term bonds, and chooses at every instant what fraction of its scheduled debt service to pay.[3] The part of the debt service it does not pay accrues as arrears and adds to the stock of debt.

Preferences and endowment. The sovereign maximizes

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t}\, u(c_t)\, dt,$$

3. We maintain the standard assumptions of centralized borrowing and centralized default, so that we do not make explicit the private sector and the policy instruments through which the sovereign implements its preferred outcomes. For the case of decentralized borrowing see Kim and Zhang (2012).

where ρ is its discount rate and u is increasing and strictly concave. Log productivity z_t follows a Markov process with generator $\mathcal{L}_z$, which we leave general in this section: in Section 3 it is constant or a finite-state chain, in Section 4 an Ornstein–Uhlenbeck diffusion. Output is e^{z_t} when the sovereign is current on its debt; when it defaults, output is $\phi(d_t, z_t)\, e^{z_t}$, where the cost function ϕ is described below.

Debt. The stock of debt is B_t. Each unit of debt pays a coupon λ and matures at rate δ, so that the scheduled debt service is $(\delta + \lambda)B_t$ per unit of time.[4] The sovereign chooses a *default intensity* $d_t \in [0, 1]$, the share of the scheduled debt service it withholds. Of every unit withheld, a fraction κ is added to the stock of debt as arrears; $1 - \kappa$ is the haircut suffered by lenders on defaulted payments. The sovereign also issues ℓ_t units of new debt at the market price q_t (when $\ell_t < 0$ it buys debt back). The budget constraint and the law of motion of debt are

$$c_t = \phi(d_t, z_t)\, e^{z_t} - (1 - d_t)(\delta + \lambda)B_t + q_t\, \ell_t, \tag{1}$$

$$\dot{B}_t = \ell_t - \delta B_t + \kappa(\delta + \lambda)\, d_t\, B_t. \tag{2}$$

The three terms in (1) are income net of the default cost, the part of the debt service that is paid, and the proceeds from issuance; the three terms in (2) are issuance, amortization and the accrual of arrears. It is convenient to write

$$a(d) := \delta - \kappa(\delta + \lambda)\, d, \qquad a := a(1) = \delta - \kappa(\delta + \lambda), \tag{3}$$

so that $\dot{B}_t = \ell_t - a(d_t)B_t$: absent issuance, debt decays at rate $a(d)$, which is lowest in full default. We assume $a > 0$, so that even a sovereign in full default sees its debt shrink, slowly, when it does not issue.

The cost of default. Following Arellano, Mateos-Planas, and Ríos-Rull (2023) the cost of default has a variable and a fixed component,

$$\phi(d, z) = \left(1 - \gamma_0 d^{\gamma_1}\right) F(z) \;\text{ for } d > 0, \qquad \phi(0, z) = 1, \qquad F(z) = 1 - \gamma_2\, (e^z - z^*)\, \mathbb{1}\{e^z \geq z^*\}, \tag{4}$$

with $\gamma_0 \in (0, 1)$, $\gamma_1 > 1$ and $\gamma_2 \geq 0$. The variable cost $\gamma_0 d^{\gamma_1}$ is convex in the default intensity with zero marginal cost at $d = 0$. The fixed cost $(1 - F(z))e^z$ is paid whenever the sovereign defaults at all, however little, and only when productivity exceeds the threshold z^*; it is larger in better times. Below the threshold default carries only the variable cost, and we call these states the *free-default region*. The jump of ϕ at $d = 0$ above the threshold is what creates a region of no default: a small default is not worth its fixed cost.

4. This is the continuous-time counterpart of the long-term bond of Hatchondo and Martinez (2009) and Chatterjee and Eyigungor (2012): the maturing fraction δ of the stock is repaid at par, and a single state variable summarizes the debt.

The drift of debt. Substituting ℓ_t out of (1) using (2), the drift of debt is a function of the state, the price and the two controls,

$$S(B,z,c,d,q) = \frac{c - \phi(d,z)e^z + (1-d)(\delta+\lambda)B}{q} - a(d)\,B, \tag{5}$$

and we treat consumption c and the default intensity d as the controls. Equivalently, for a given drift S,

$$c = \mathcal{R}(d;B,z,q) + q\,S, \qquad \mathcal{R}(d;B,z,q) := \phi(d,z)e^z - (1-d)(\delta+\lambda)B + q\,a(d)\,B. \tag{6}$$

We call $\mathcal{R}$ the *resource function*: it is consumption at zero drift, and it is all that the default decision affects for a given drift. The characterization of Section 3 rests on this decomposition.

Arrears as borrowing. The decomposition has a second reading that is the first of the paper's three claims. Let $n_t := \ell_t + \kappa(\delta+\lambda)d_t B_t$ be the total addition to the stock of debt, from issuance and from arrears, so that $\dot{B}_t = n_t - \delta B_t$. Substituting into (1),

$$c_t = \phi(d_t,z_t)e^{z_t} - (\delta+\lambda)B_t + q_t\,n_t + (1-\kappa q_t)(\delta+\lambda)B_t\,d_t. \tag{7}$$

Holding the path of debt fixed, withholding a unit of debt service raises consumption by $1 - \kappa q_t > 0$: the unit saved, net of the market value of the κ units of arrears that replace market issuance one for one. Equivalently, arrears are the issuance of κ units of debt to existing creditors at the implicit price $1/\kappa$, which exceeds the market price and does not depend on it, paid for with the output cost $(1-\phi)e^z$ of default. Section 3.1 draws the consequences.

The HJB equation. Given a bond price function $q(B,z)$, the sovereign's value $V(B,z)$ solves

$$\rho V(B,z) = \max_{c,\ d\in[0,1]} \Big\{ u(c) + S(B,z,c,d,q(B,z))\,V_B(B,z) + \mathcal{L}_z V(B,z) \Big\}, \tag{8}$$

subject to the constraints on issuance described in Section 2.4 (the state constraint $B \geq 0$, and the underwriting standard). Subscripts denote partial derivatives. The derivation is in Appendix A. Unlike models with a binary default decision, there is no variational inequality and no second regime: the sovereign never loses market access, and a single value function with possibly discontinuous policies describes it at all states.

First-order conditions. At an interior drift, consumption satisfies

$$u'(c)\,q(B,z) = -V_B(B,z), \tag{9}$$

the price of a bond times the marginal utility of the resources it brings equals the marginal cost of debt in utility units. By (6) the default intensity enters the objective only through $\mathcal{R}$, which is

strictly concave on $(0, 1]$ with

$$\mathcal{R}_d(d) = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1} + (1 - \kappa q)(\delta + \lambda)B.$$

The marginal gain from withholding a unit of debt service is $(1 - \kappa q)(\delta + \lambda)B$: the payment saved, net of the market value κq of the arrears it creates. The interior maximizer is

$$d_{\text{int}}(B, z, q) = \min\Big\{1,\ \Big[\frac{(1 - \kappa q)(\delta + \lambda)B}{\gamma_0\gamma_1 F(z)e^z}\Big]^{1/(\gamma_1 - 1)}\Big\}, \tag{10}$$

and the optimal default intensity is either 0 or d_{int}, whichever gives the larger resource function. Section 3.1 makes this precise and shows that it holds for every maximizer of (8).

2.2 International Lenders

A unit measure of deep-pocketed, competitive, risk-neutral lenders with outside option r buy and hold the sovereign's bonds. A unit of debt held at t yields the flow $(1 - d_t)(\delta + \lambda)$ in cash and $\kappa(\delta + \lambda)d_t$ in new units (the arrears), and δ of it matures. The break-even price is the expected discounted value of the cash flow, where the discount rate is adjusted for the growth of the position through arrears,

$$q(B, z) = \mathbb{E}_{B,z} \int_0^\infty \exp\Big(-\int_0^t \xi(d_s)\, ds\Big)\,(1 - d_t)(\lambda + \delta)\, dt, \qquad \xi(d) := r + \delta - \kappa(\delta + \lambda)\, d. \tag{11}$$

The effective discount rate $\xi(d)$ combines the lenders' opportunity cost r, the amortization δ, and the accrual of arrears, which lowers it. By the Feynman–Kac formula, q solves the linear equation

$$\xi\big(d(B, z)\big)\, q(B, z) = (1 - d(B, z))(\lambda + \delta) + \tilde{S}(B, z)\, q_B(B, z) + \mathcal{L}_z q(B, z), \tag{12}$$

where $\tilde{S}(B, z)$ is the equilibrium drift of debt and $d(B, z)$ the equilibrium default policy. The evolution of the stock of debt is governed by $\tilde{S}$, which includes issuance; the cash flow of a single unit is governed by $\xi(d)$, which does not.

Spreads and duration. The risk-free price is $\bar{q} = (\delta + \lambda)/(r + \delta)$, which we normalize to 1 by setting $\lambda = r$. The yield to maturity implied by a price q is $(\delta + \lambda)/q - \delta$, so the spread over the risk-free yield is

$$\text{spread}(B, z) = \frac{\delta + \lambda}{q(B, z)} - (\delta + r), \tag{13}$$

and the risk-free Macaulay duration of the bond is $1/(\delta + r)$.

2.3 Equilibrium and the Ergodic Set

Definition 1. A *Markov equilibrium* consists of a value function $V(B,z)$, policies $c(B,z)$ and $d(B,z)$, and a bond price $q(B,z)$ such that

1. given q, the value V solves the HJB equation (8) and (c,d) attain the maximum on its right-hand side, subject to the state constraint $B \geq 0$ and the underwriting standard of Section 2.4;
2. given the policies, q solves the pricing equation (12), equivalently (11).

The stationary distribution. Let $g(B,z)$ be the density of the state. Given the equilibrium drift $\tilde{S}$ and the generator $\mathcal{L}_z$ with adjoint $\mathcal{L}_z^*$, the density solves the Kolmogorov forward equation

$$\partial_t g = -\partial_B \left[\tilde{S}(B,z)\, g\right] + \mathcal{L}_z^* g, \tag{14}$$

and the stationary distribution solves $0 = -\partial_B[\tilde{S}g] + \mathcal{L}_z^* g$. The forward equation is the adjoint of the HJB equation: in the discretized model the same sparse matrix that solves the HJB equation, transposed, gives the stationary distribution, so that the long-run moments are available without simulation (Appendix B). The stationary distribution can have atoms: at $B = 0$ when the sovereign repays in full at high productivity, and at stationary points of debt in the deterministic and chain cases. We call its support the *ergodic set*; its shape, in particular how far it extends into the default region, is a central object of the paper.

2.4 Discussion

Comparison with the discrete-time model. The model follows Arellano, Mateos-Planas, and Ríos-Rull (2023) in letting the sovereign withhold a share of the scheduled payments at a cost that is convex in the share, with a fixed cost that depends on productivity. Three points of comparison are worth noting. First, the discrete-time model distinguishes between the debt due in the period and the stock of debt; in continuous time the debt due is the flow $(\delta + \lambda)B$ and the two coincide up to the factor $\delta + \lambda$. Second, the sovereign in our model can buy back debt ($\ell < 0$); this matters for the monotonicity of the value in debt, which we prove with buybacks allowed. Third, as in their model but unlike the standard model with a binary default decision, the sovereign never loses market access: the price at which it can issue falls with the expected default intensity, which is the only punishment through markets. There is therefore a single value function, no stopping problem, and no variational inequality, which is what makes the default decision static (Theorem 3) and the characterization of Section 3 possible.

The underwriting standard. Debt is restricted to $B \geq 0$ (the sovereign does not lend) and, in the computation, to $B \leq \bar{B}$. Without a further restriction the upper limit is not a harmless truncation: in the states where default carries no fixed cost, a sovereign in full default faces almost no marginal cost of debt while lenders still pay a positive price, because productivity will

recover, so the sovereign issues at any positive price and debt grows without bound (Section 3.2). We therefore impose a restriction on lenders: they do not absorb new issues at a price below a floor $\underline{q}$,

$$\ell_t \leq 0 \quad \text{whenever} \quad q(B_t, z_t) < \underline{q}. \tag{15}$$

Buybacks remain allowed, and partial default and the accrual of arrears are unchanged. We call (15) an *underwriting standard*, after the seniority and underwriting arrangements of Chatterjee and Eyigungor (2015). In yield terms, with $\lambda = r$ the floor corresponds to a spread of $(\delta + \lambda)(1/\underline{q} - 1)$; for the value $\underline{q} = 0.7$ we use, this is about 900 basis points at the calibrated maturity rate (700 at the maturity rate of the calibration without the standard, at which the level was chosen). The level is a judgment: it clears the highest price at which the unconstrained sovereign still issues in full default with a margin, and binds on ordinary partial-default borrowing only in deep distress (Appendix C). Section 3.2 shows what the standard does: it makes the full-default region transient, the price continuous and the support of debt independent of the grid, and where it binds it makes the sovereign apply its default rule at a shadow price below the market price. It binds only in deep distress: at the calibrated parameters, on about two percent of the ergodic mass, in the lower third of the productivity range.

No trade at a zero price. A second, milder convention rules out the autarky equilibrium in which lenders expect full default forever, the price is zero, and the sovereign keeps itself above the full-default line by issuing at a zero price. We impose that no bonds change hands at a zero price; Remark 8 shows that the price is then strictly positive at every state.

3 Characterization

This section collects what can be established analytically, and it formalizes the paper's three claims: arrears as borrowing and its consequences in Section 3.1, the behavior at high debt and the underwriting standard in Section 3.2. Section 3.1 contains results that hold for any productivity process, and Section 3.2 the behavior at high debt and what the underwriting standard does; Sections 3.3 and 3.4 specialize to constant productivity and to a finite-state chain, each with a numerical illustration at the parameters of the calibrated diffusion model of Section 4. Proofs are in Appendix A. Throughout we maintain the following.

Assumption 1.

(i) $\xi(1) = r + \delta - \kappa(\delta + \lambda) > 0$ and $a = \delta - \kappa(\delta + \lambda) > 0$.

(ii) $\lambda = r$, so that $\bar{q} = 1$; $\rho > r$.

(iii) u is C^2, strictly increasing and strictly concave; $\gamma_1 > 1$ and $0 < \gamma_0 < 1$.

(iv) The drift is bounded, $|S| \leq \bar{S}$, and $B \in [0, \bar{B}]$.

3.1 Results for any productivity process

Proposition 2 (arrears are borrowing at a fixed price)**.** *For any path of total new debt n_t, consumption satisfies (7): a unit of debt service withheld is κ units of debt issued to existing creditors at the price $1/\kappa$, independent of the market price, in exchange for the output cost of default. In particular, for a given drift of debt the default intensity affects consumption only through the resource function $\mathcal{R}(d; B, z, q) = \phi(d, z)e^z - (1-d)(\delta+\lambda)B + q\,a(d)B$, and $\mathcal{R}_d = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1} + (1-\kappa q)(\delta+\lambda)B$ for $d > 0$.*

The proposition is an identity, but it is the identity that organizes the model: the sovereign has two ways of raising funds, the market at the price q and arrears at the price $1/\kappa$, and the second is always the cheaper per unit of debt and always costs output. Everything below follows from it, starting with the fact that the choice between the two is static.

Theorem 3 (the default decision is static)**.** *For every state (B, z), price $q \geq 0$ and value function V, every maximizer (d^*, c^*) of the right-hand side of (8) satisfies*

$$d^* \in \arg\max_{d \in [0,1]} \ \mathcal{R}(d; B, z, q).$$

The default policy depends on $(B, z, q(B, z))$ only, and not on V, on the sign of the drift, or on the constraints on the drift, as long as those constraints do not depend on d.

The qualification matters for the underwriting standard, whose constraint $\ell = S + a(d)B \leq 0$ does depend on d; Proposition 10 in Section 3.2 gives the rule in that case. Everywhere else the default decision is a static maximization of the concave function $\mathcal{R}$, which yields explicit thresholds.

Proposition 4 (default thresholds)**.** *Fix (B, z, q) with $B > 0$ and $\kappa q < 1$. The optimal default intensity is $d^* = 0$ if the fixed cost exceeds the gain from defaulting,*

$$(1 - F(z))\, e^z \ \geq \ \Delta(B, z, q) := (1-\kappa q)(\delta+\lambda)B\, d_{\text{int}} - \gamma_0 F(z) e^z d_{\text{int}}^{\gamma_1}, \tag{16}$$

and $d^* = d_{\text{int}}(B, z, q)$ of (10) otherwise. Δ is continuous, strictly increasing in B, strictly decreasing in q, and zero at $B = 0$. Consequently:

(a) there is a threshold $B_{\text{pd}}(z, q)$, nondecreasing in q, with $d^* = 0$ for $B \leq B_{\text{pd}}$ and $d^* = d_{\text{int}} > 0$ for $B > B_{\text{pd}}$;

(b) in the free-default region ($F(z) = 1$), $B_{\text{pd}}(z, q) = 0$: any positive debt is partially defaulted on;

(c) if $F(z) < 1$ and $d_{\text{first}}(z) < 1$ (below), the default intensity jumps at B_{pd} from 0 to the first default intensity

$$d_{\text{first}}(z) = \Big[\frac{1 - F(z)}{(\gamma_1 - 1)\gamma_0 F(z)}\Big]^{1/\gamma_1}, \tag{17}$$

which depends on the cost parameters only;

(d) full default, $d^* = 1$*, occurs for* $B \geq \max\{B_{\text{fd}}(z,q), B_{\text{pd}}(z,q)\}$ *with* $B_{\text{fd}}(z,q) := \gamma_0\gamma_1 F(z)e^z / \big((1-\kappa q)(\delta+\lambda)\big)$*; when* $d_{\text{first}}(z) \geq 1$ *the intensity jumps from* 0 *directly to* 1 *at* B_{pd}*;*

(e) when $d_{\text{first}} < 1$*, the threshold has the closed form*

$$B_{\text{pd}}(z,q)^{\frac{\gamma_1}{\gamma_1-1}} = \frac{\gamma_1}{\gamma_1-1}\,\frac{(1-F(z))\,e^z\,G^{\frac{1}{\gamma_1-1}}}{K^{\frac{\gamma_1}{\gamma_1-1}}}, \qquad G := \gamma_0\gamma_1 F(z)e^z, \quad K := (1-\kappa q)(\delta+\lambda). \tag{18}$$

Corollary 5 (monotonicity of the default policy)**.** $d^*(B,z,q)$ *is nondecreasing in* B *and nonincreasing in* q*, in* γ_0*, in* κ*, and in* $F(z)e^z$ *holding the fixed cost* $(1-F(z))e^z$ *fixed. The thresholds* B_{pd} *and* B_{fd} *are increasing in* γ_0*, in* κ *and in* q*, and decreasing in* $\delta+\lambda$*; when* $d_{\text{first}} < 1$*,* B_{pd} *is increasing in the fixed cost* $1-F(z)$*.*

The thresholds are stated at a given price; in equilibrium the no-default region is $\{B \leq B_{\text{pd}}(z, q(B,z))\}$, a fixed-point condition, which is explicit at the risk-free price. Two implications follow directly, and both are testable in principle. First, a sovereign never defaults a little: the first default is the jump d_{first}, which depends on the cost parameters only, and at the calibrated parameters it is half of the debt service, $d_{\text{first}} \approx 0.51$ at average productivity, because the fixed cost is small relative to the variable cost. Second, at the risk-free price the no-default region at average productivity ends at $B_{\text{pd}}(0,1) = 0.456$, which is where the deterministic steady state sits (Theorem 14) and where the ergodic distribution of the diffusion model concentrates (Section 4.3): the debt limit of the partial default model is an explicit function of the primitives.

Proposition 6 (the bond price)**.** *Fix measurable policies* (d,S) *and let* (B_t,z_t) *be the induced process. The bounded solution of* (12) *is* (11)*, and it satisfies* $0 \leq q \leq \bar{q}$*. Moreover* $q(B,z) = \bar{q}$ *if and only if the sovereign never defaults along the path from* (B,z)*, and* $q(B,z) = 0$ *if and only if it is in full default forever.*

Proposition 7 (continuity of the price along crossing flows)**.** *Fix a state* (B_0,z_0) *and suppose that on a neighborhood of it the equilibrium drift is negative and bounded away from zero, so that from every nearby state debt moves down through* B_0 *in a time that vanishes with the distance to* B_0*. Then* $q(\cdot,z_0)$ *is continuous at* B_0*: with no further condition in the deterministic and chain models, and whenever* $q(B_0,\cdot)$ *is continuous at* z_0 *in the diffusion model. The same holds with positive drift. Hence* q *can be discontinuous in* B *only at zeros of the drift that are not approached from both sides.*

Proposition 7 is the mechanism of the price cliffs of Section 3.2: at a stationary point of debt that is approached from below and left from above, the paths on the two sides go to different places and the price jumps. In an equilibrium whose stationary points are all approached from both sides the price is continuous everywhere.

Remark 8 (autarky and its removal). Under a zero price, full default is optimal if and only if $B \geq B_{\text{fd}}(z,0)$, and $(q \equiv 0, d \equiv 1)$ is an equilibrium on any invariant set contained in $\{B \geq B_{\text{fd}}(z,0)\}$; invariance requires the sovereign to issue at least aB at a zero price. Under the convention that no bonds are traded at a zero price, $q > 0$ at every state in every equilibrium.

Finally, the value is monotone in debt, with buybacks and with the standard; Proposition 10 in Section 3.2 uses this for the sign of V_B.

Proposition 9 (the value is nonincreasing in debt)**.** *If for every z the price $q(\cdot, z)$ is nonincreasing and strictly positive, and the drift cap is slack along the optimal path and along the mimicking path of the proof, then $V(\cdot, z)$ is nonincreasing, with or without buybacks and with or without the underwriting standard.*

3.2 High Debt and the Underwriting Standard

This subsection formalizes the paper's second and third claims. Two things happen in the model at high debt, and the discrete-time literature encounters them together with a third, numerical one. The first is the knife-edge autarky equilibrium of Remark 8, which a convention removes. The second is a genuine explosion of debt, a property of the equilibrium whose entry point looks like a second steady state in computed solutions; we describe it here and remove it with the underwriting standard, which is the third claim. The third, a resolution artifact of first-order schemes that mimics a second basin of attraction, is described in Appendix B.3. The numerical evidence behind this subsection, at the parameters of the first of the two calibrations of Section 4.2, is in Appendix C; at those parameters the fixed-cost threshold sits well inside the productivity range. At the final parameters of Table 1 it sits at the lower edge of the grid, so the explosion is latent there and the standard binds in partial default only (Section 4.3).

The free-default explosion. The explosion is a property of the equilibrium, not of the discretization. In the free-default region a sovereign in full default pays nothing, bears only the variable cost, and faces a marginal cost of debt $-V_B$ close to zero, while the price is positive because productivity will recover; by (9) it issues at any positive price, at a rate of a third to a half of output per year at the parameters of Appendix C, up to the top of the grid. The entry point is a stationary point approached from below (the sovereign borrows up to a self-imposed limit just below the full-default line) and left from above, and the price jumps there by Proposition 7. On the cliff nodes the drift sign alternates between iterations, the residual of the pricing equation stays large and grows under refinement, and the share of the ergodic mass that escapes through the cliff is decided at the scale of the grid: it survives every refinement and moves the mean partial default and the spread volatility by tens of percent across grids (Appendix C, Table 3). No refinement fixes this.

The underwriting standard. With issuance switched off below $\underline{q}$, the drift beyond the full-default line is $-a(d)B < 0$, the full-default boundary is approached from both sides, the price is continuous across it, the chatter disappears, and the top of the grid is an exact truncation (Proposition 11). The floor must exceed the highest price at which the unconstrained sovereign still issues in full default; a lower floor does not remove the accumulation but relocates it to the debt level at which the price falls through the floor (Appendix C, Table 4). We set $\underline{q} = 0.7$,

which clears that cutoff with a margin and binds on ordinary partial-default borrowing only in the lowest productivity states; with it the price residual converges.

What the standard does. Three results describe the equilibrium with the standard in place. Its constraint $\ell \leq 0$ depends on d through $\ell = S + a(d)B$, so the separation of Theorem 3 must be re-examined where it binds.

Proposition 10 (the default decision under the standard)**.** *Fix a state with $q = q(B,z) < \underline{q}$, $B > 0$ and $V_B < 0$, and let (d^*, S^*) be the unconstrained maximizer with issuance $\ell^* = S^* + a(d^*)B$.*

(a) If $\ell^* \leq 0$ the standard is slack and Theorem 3 holds verbatim.

(b) If $\ell^* > 0$ and $d^* > 0$ (the unconstrained optimum is in partial default) the constrained maximum has $\ell = 0$, hence $S = -a(d)B$, and the default intensity maximizes

$$J(d) := u\big(\phi(d,z)e^z - (1-d)(\delta+\lambda)B\big) - a(d)\,B\,V_B.$$

At an interior maximizer d^{**},

$$d^{**} = d_{\text{int}}(B,z,\hat{q}), \qquad \hat{q} := \frac{-V_B}{u'(c^{**})} \in (0,q) :$$

the unconstrained rule (10) evaluated at the sovereign's own shadow price of a unit of debt instead of the market price. Since d_{int} is decreasing in the price, $d^{**} > d_{\text{int}}(B,z,q)$.

Where the standard binds, a unit of arrears is valued by the sovereign at $\hat{q}$, which is below the market price because the sovereign would like to issue and cannot, and the sovereign defaults more than the market-price rule prescribes. The numerical kernel maximizes over (d,S) jointly and therefore implements the constrained rule; at the parameters of Table 1 the standard binds in partial default on about two percent of the ergodic mass, where the default intensity exceeds the market-price rule by four percent of that rule on average and by up to eight percent.

Proposition 11 (the standard removes the explosion)**.** *Let $\underline{q}$ exceed the price at every state in which the sovereign is in full default. Then in every equilibrium:*

(a) in full default the drift is $S \leq -aB < 0$: the full-default region is transient, and in the deterministic model it is left in time at most $a^{-1}\ln(B/B_{\text{fd}})$;

(b) in the deterministic model the drift has no zero in $\{q < \underline{q}\}$; a stationary point on the floor locus $\{q = \underline{q}\}$ is approached from above by forced repayment and from below by borrowing, and the price is continuous across it whenever it is reached in finite time from both sides;

(c) any $\bar{B}$ above the full-default thresholds is an exact truncation: the drift points inward at $\bar{B}$ in every productivity state and the stationary distribution puts no mass on it;

(d) the no-default region, the thresholds of Proposition 4 and the deterministic steady state of Theorem 14 are unaffected whenever $\underline{q} < 1$.

Proposition 12 (the standard bounds the support of debt)**.** *Let $q > 0$ everywhere and $B^\sharp := \sup\{B : q(B,z) \geq \underline{q}$ for some $z\}$. Then in every equilibrium with any $\bar{B} \geq B^\sharp$: whenever $B_t > B^\sharp$ the drift satisfies $S_t \leq -aB_t$, the set $[0, B^\sharp] \times Z$ is invariant, every stationary distribution is supported in it, and it does not depend on $\bar{B}$.*

Propositions 11 and 12 address different failures: the first the explosion inside the default region, the second the dependence on the grid. At the parameters of Table 1 the second is slack ($B^\sharp$ exceeds the grid, since the price at the top of the grid in the best productivity states is still above the floor); what keeps the mass off the top of the grid there is the inward drift the sovereign chooses at high debt in every state, which the first proposition guarantees in full default.

3.3 The Deterministic Case

With z fixed, normalized to 0, the equilibrium is a pair of ordinary differential equations. Writing S^*, c^*, d^* for the policies, (8) and (12) read

$$V'(B) = \frac{\rho V(B) - u(c^*)}{S^*}, \qquad q'(B) = \frac{\xi(d^*)q(B) - (1-d^*)(\lambda+\delta)}{S^*}, \tag{19}$$

with $d^* = d^*(B, q(B))$ from Proposition 4, $c^* = (u')^{-1}(-V'/q)$ and $S^* = (c^* - \mathcal{R}^*)/q$. The system is singular where the drift vanishes, and the stationary points of debt organize the solution. We call a zero of the drift a *steady state*. The theory pins down one of them, the no-default steady state B^* of Theorem 14; a zero inside the partial-default region would be a self-fulfilling partial-default steady state, discussed at the end of this subsection; and in the chain model a steady state at the risk-free price can also sit below the default threshold, held in place by precaution (Section 3.4).

Definition 13. An equilibrium is of *no-default-basin type* with steady state $B^* \in (0, \bar{B})$ if $d \equiv 0$ on $[0, B^*]$, $S > 0$ on $[0, B^*)$, $S(B^*) = 0$ and $S < 0$ on $(B^*, \bar{B}]$. It satisfies the *finite-reach condition* if from $B^* + \eta$ the state reaches B^* in a time $\tau(\eta) \to 0$ as $\eta \downarrow 0$.

Theorem 14 (the deterministic steady state in closed form)**.** *Let (V, q) be a no-default-basin equilibrium satisfying the finite-reach condition. Then $q \equiv 1$ on $[0, B^*]$, $V(B^*) = u(1 - \lambda B^*)/\rho$, and*

$$B^* = B_{\mathrm{pd}}(0, 1),$$

the largest debt the sovereign is willing to service in full at the risk-free price, given explicitly by (18) with $q = 1$.

The steady state is where the impatient sovereign, borrowing at the risk-free price, runs into its own default threshold: it is the debt limit of the partial default model, the counterpart of the

limit that the binary model pins down only numerically, here in closed form. It is increasing in the default costs γ_0 and $1-F$, in the recovery κ and in income, decreasing in $\delta+\lambda$, and independent of ρ (as long as $\rho > r$), of the curvature of u and of the drift cap. Steady-state consumption is $1-\lambda B^*$ and the first partial default beyond B^* has intensity d_{first}. On the basin $[0, B^*)$ the sovereign faces a constant price of one and borrows up to the steady state, which it reaches in finite time; the approach is governed by a square-root law.

Remark 15 (the approach from the left). On the basin $q \equiv 1$ and $d \equiv 0$, so $c = 1-\lambda B+S$ and the HJB equation reads $\rho V = u(c) + SV'$ with $V' = -u'(c)$. Differentiating in B and substituting the first-order condition gives $SS' = \lambda S - (\rho-\lambda)\,\varphi(c)$ with $\varphi := u'/|u''|$. As $B \uparrow B^*$ the drift vanishes, so $(S^2)' \to -2(\rho-\lambda)\varphi(\bar{c})$ and

$$S(B^*-x) = \big(2(\rho-\lambda)\,\varphi(\bar{c})\big)^{1/2} x^{1/2}\,(1+o(1)).$$

The time to reach B^* from B^*-x is $\int_0^x dy/S(B^*-y) \propto x^{1/2}$, which is finite: the finite-reach condition holds from the left. The square-root law is the impatient sovereign's approach to a limit it cannot cross at the risk-free price; Theorem 16 gives the approach from the other side.

Beyond B^* the equilibrium is more delicate: the sovereign is in partial default, the price is below one, and the drift is negative, so the state returns to B^* from above. The next result describes the approach.

Theorem 16 (the cube-root law at the steady state)**.** *Let (V,q) be a no-default-basin equilibrium with the finite-reach condition and $d_{\text{first}} < 1$, and suppose that on a right neighborhood of B^* the drift is interior and V is C^1; let u be C^3 near $\bar{c}$ (for the second-order terms) and real-analytic near $\bar{c}$, as CRRA utility is (for the last statement). Write $x = B-B^*$, $\bar{c} = 1-\lambda B^*$, and*

$$\Pi_0 := (1-\kappa)(\delta+\lambda)\,d_{\text{first}}, \qquad \beta := \Big(3\,a(d_{\text{first}})\,B^*\,\Pi_0\,\frac{u'(\bar{c})}{|u''(\bar{c})|}\Big)^{1/3}, \qquad \alpha := \frac{3\Pi_0}{2\beta}.$$

Then as $x \downarrow 0$,

$$S(B^*+x) = -\beta\,x^{1/3}\,(1+o(1)), \qquad c(B^*+x) = \bar{c} - \beta\,x^{1/3}\,(1+o(1)),$$
$$1-q(B^*+x) = \alpha\,x^{2/3}\,(1+o(1)),$$

and $V'(B^{*+}) = -u'(\bar{c}) = V'(B^{*-})$. More precisely, with $\varphi(c) := u'(c)/|u''(c)|$ and $R_B := -(1-d_{\text{first}})(\delta+\lambda) + a(d_{\text{first}})$,

$$S(B^*+x) = -\beta\,x^{1/3} - \Sigma_1\,x^{2/3} + O(x), \qquad 1-q(B^*+x) = \alpha\,x^{2/3} + Q_1\,x + O(x^{4/3}),$$
$$\Sigma_1 = -\frac{1}{4}\Big[2a(d_{\text{first}})B^*\alpha + \beta^2\frac{\varphi'(\bar{c})}{\varphi(\bar{c})} + \frac{3\varphi(\bar{c})\,(R_B+\rho)}{\beta}\Big], \qquad Q_1 = -\frac{\Pi_0\,\Sigma_1}{\beta^2},$$

where $\varphi'(\bar{c})/\varphi(\bar{c}) = 1/\bar{c}$ for CRRA utility. Thus V is C^1 but not C^2 at the steady state, the drift leaves zero with a vertical tangent, and the price has a vertical tangent from the right. Moreover the right

branch exists on a neighborhood of B^*, is unique among branches that stay bounded in the scaled variables $(1-q)/x^{2/3}$ and $-qS/x^{1/3}$ as $x \downarrow 0$ (hence among those with these asymptotics), and is an analytic function of $(B-B^*)^{1/3}$ (Appendix A).

The economics of the law is that the debt limit is a soft edge with a sharp price: a sovereign just over its limit does not snap back, it repays slowly at first and faster as it moves away, while even a small excess of debt carries a visibly lower price; partial default is what makes this possible, since with a binary decision there would be a jump instead. The law has a simple origin. On the right branch the HJB equation reads $\rho V = u(c) + w\,u'(c)$ with $w = -qS \geq 0$ the resources devoted to repayment; letting $B \downarrow B^*$ forces $c \to \bar{c}$ and hence $S \to 0$, and differentiating gives $w' \approx K/w^2$, whence $w^3 \approx 3Kx$. The price gap then integrates $q' \approx -\Pi_0/w$ to the two-thirds power. The constant Π_0 is the flow of lost coupons at the first default; the price has a vertical tangent on the right of the steady state because a small excursion into default costs lenders a flow of coupons over a time that is only $O(x^{2/3})$ but at a rate that does not vanish. Numerically, this is why one-sided differences of V on a production grid look like a kink: on a grid of step h the right difference averages V' over $[B^*, B^*+h]$ and differs from $-u'(\bar{c})$ by $O(h^{1/3})$.

Theorem 17 (existence of the right branch)**.** *Under the hypotheses of Theorem 16 and the underwriting standard with $\underline{q}$ above the full-default prices, suppose the maximal solution of (19) started along the cube-root branch keeps $S^* < 0$, $c^* > c_{\min}$ and $|S^*| < \bar{S}$ up to the full-default threshold, and that beyond it buybacks are not optimal (the corner condition $qu'(c) \leq -V_B$ holds, as it does at the computed solutions, where $-V_B$ is small in full default). Then the right branch exists on all of $[B^*, \bar{B}]$. Where the standard binds before full default, on $\{q < \underline{q},\ \ell^* > 0\}$, the system switches to the constrained regime of Proposition 10(b), $S = -a(d^{**})B$ with d^{**} at the shadow price, and consumption steps down at the switch; beyond the full-default threshold the sovereign neither issues nor buys back, debt decays as $B_t = B_0 e^{-at}$, the price is the power $q(B) \propto B^{-\xi(1)/a}$, and V is given by quadrature. The branch is unique among those with the cube-root asymptotics, q is continuous on $[0, \bar{B}]$, and V is C^1 below the point where the standard starts to bind.*

The condition is that the drift has no second zero before full default, that is, no steady state in partial default. It holds at the computed solutions, but it is a property of the solution, not of the primitives, and a second zero would be a self-fulfilling partial-default steady state: a pessimistic price sustains a default intensity that sustains the price.

Lemma 20 in Appendix A gives the local conditions at such a point: the stationary price is what lenders pay for a constant default intensity, the slope of the price is pinned by the impatience wedge, $-q'/q = (\rho - r)/(a(d^\circ)B^\circ)$, and the slope of the drift then follows from the pricing equation, so that the point attracts only if additional debt dilutes the price strongly enough through additional default. These conditions do not select: they admit a continuum of locally consistent candidates, one at every debt level in an interval above the no-default steady state. Whether any of them extends to a global equilibrium is a two-point problem: the sovereign must be willing to borrow into partial default, at prices below one, all the way from zero debt

to the candidate. At the computed solutions none does, for a reason that we can state precisely: with an impatience wedge of a few dozen basis points, the sovereign facing the candidate's prices prefers to repay out of the default region rather than borrow into it at the spread the candidate implies, which breaks the candidate. At larger wedges the marching algorithm of Appendix B.1 finds candidates that pass the two-point test, and we leave their characterization to future work. We record the general case as a conjecture; it is the continuous-time form of the slow-moving crises of Lorenzoni and Werning (2019), and we return to it in the conclusion.

Conjecture 18. *A self-fulfilling partial-default steady state exists if and only if the impatience wedge $\rho - r$ exceeds the spread implied by the first default intensity over the expected length of the default spell. At the calibrated parameters the no-default-basin equilibrium is the unique equilibrium with $q > 0$ everywhere and q continuous.*

Numerical illustration. Figure 1 shows the deterministic equilibrium at the parameters of the calibrated diffusion model (Table 1), with productivity fixed at its mean and the underwriting standard at $\underline{q} = 0.7$, computed by the implicit upwind time stepping of Appendix B on a grid with 500 nodes per unit of debt, to residuals of 10^{-7} and 10^{-9}. The steady state is at the node 0.450, against $B_{\mathrm{pd}}(0,1) = 0.456$ from (18), and the first default node, 0.454, carries $d = 0.510$ against $d_{\mathrm{first}} = 0.51$ from (17). Below the steady state the price is one and the sovereign borrows; above it the sovereign is in partial default, repays, and the price falls. The underwriting standard binds from $B = 0.752$, where the price falls through 0.7: beyond that point issuance is zero, the drift is $-a(d)B$, and from $B = 1.17$, reached on a wider grid, default is complete and the price follows the power law of Theorem 17. Figure 2 shows the value and consumption: the value is smooth through the steady state, as Theorem 16 says, while consumption drops at the first default by the fixed cost and the lost issuance revenue. Figure 3 checks Theorem 16: on log-log axes the computed drift and price gap on the right of the steady state, and the drift on its left, are compared with the two-term expansions of the theorem and with the square-root law, with the constants $\beta = 0.151$, $\alpha = 0.46$, $\Sigma_1 = -0.176$ and $Q_1 = 0.361$ computed from the primitives. The leading terms alone describe the solution only within a few grid steps of the steady state: Σ_1/β is -1.2, so the second term of the drift is half the first already at $x = 0.1$, and it is the impatience term $\varphi(\bar{c})(R_B + \rho)/\beta$ that makes it large, since the first default is half of the debt service here. With two terms the drift is matched to within five percent for x up to 0.05 and the price gap to within two percent for x between 0.02 and 0.3; further out the next terms of the series take over, and within one or two grid steps of the steady state the comparison is limited by the grid's placement of the steady state, which the law itself says is off by $O(h)$. The computed drift and price gap leave the steady state with the vertical tangents the theorem predicts.

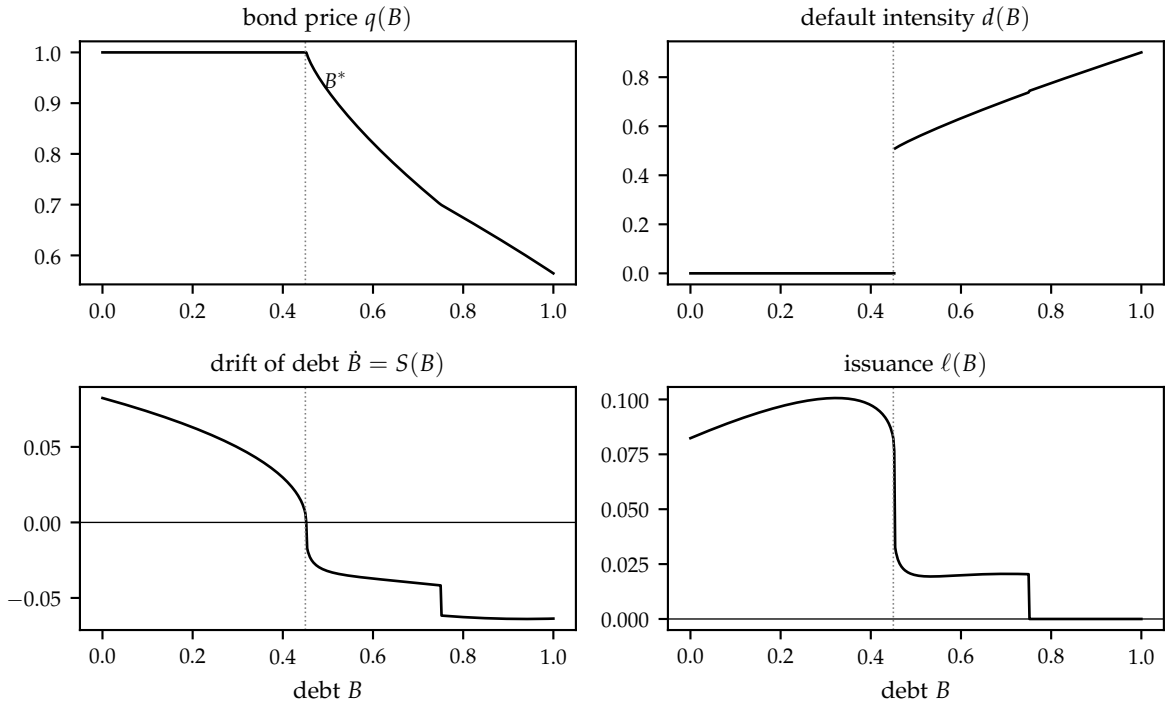


Figure 1: The deterministic case: bond price, default intensity, drift of debt and issuance as functions of debt. Calibrated parameters with productivity at its mean; dotted vertical line: the steady state $B^* = B_{\text{pd}}(0, 1)$.

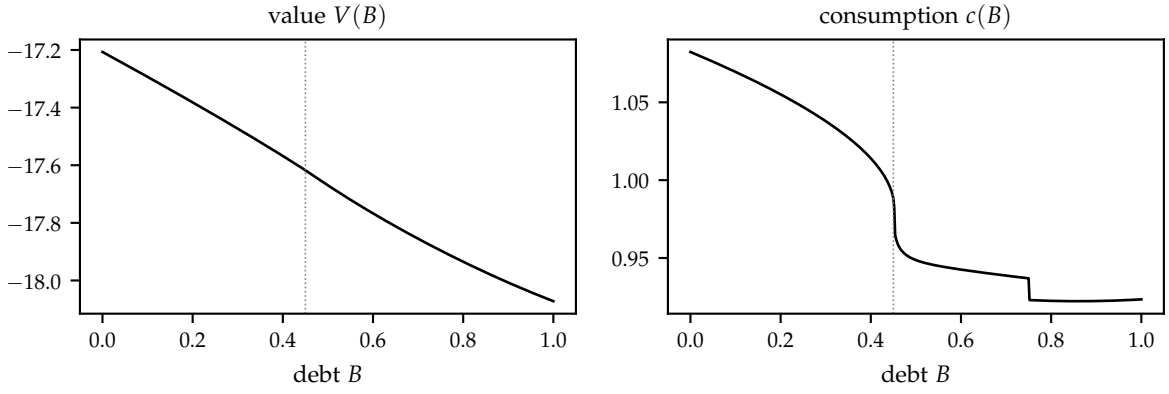


Figure 2: The deterministic case: value and consumption as functions of debt. The value is C^1 at the steady state (Theorem 16); consumption falls by the fixed cost at the first default and is nearly flat in the default region, with a small step where the underwriting standard starts to bind.

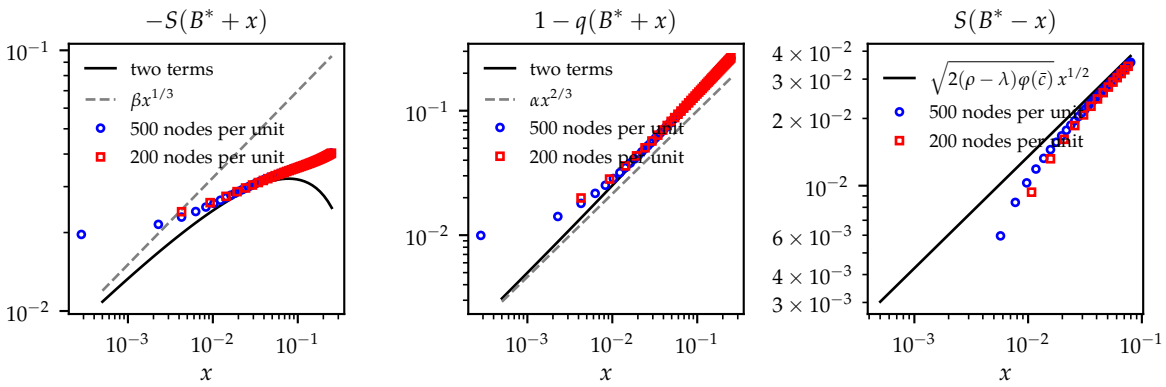


Figure 3: The cube-root law at the steady state. Markers: the computed drift and price gap at a distance x above the steady state (left and center) and the drift at a distance x below it (right), on two grids. Lines: the two-term expansions of Theorem 16 (solid) and their leading terms (dashed), with constants computed from the primitives; right panel, the square-root law.

3.4 The Jump Process Case

Let productivity take finitely many values z_j and switch from j to k at rate η_{jk}. The HJB and pricing equations are

$$\rho V_j(B) = \max_{c,d} \Big\{ u(c) + S\, V_j'(B) + \sum_{k\neq j} \eta_{jk}\big[V_k(B) - V_j(B)\big] \Big\}, \tag{20}$$

$$\xi(d_j)\, q_j(B) = (1-d_j)(\lambda+\delta) + \tilde{S}_j\, q_j'(B) + \sum_{k\neq j} \eta_{jk}\big[q_k(B) - q_j(B)\big]. \tag{21}$$

The separation theorem and the thresholds of Proposition 4 hold state by state with $q = q_j(B)$. The basin argument of Theorem 14 generalizes with one change: a steady state at the risk-free price can now be interior, held in place by the precautionary motive.

Proposition 19 (price-one steady states in the chain model)**.** *Suppose $[0, \underline{B}]$ is invariant under the equilibrium dynamics in every state and default-free. Then $q_j \equiv 1$ on $[0, \underline{B}]$ for every j and $\underline{B} \leq \min_j B_{\mathrm{pd}}(z_j, 1)$. If state j has a steady state at an interior point $B^\circ < \min_k B_{\mathrm{pd}}(z_k, 1)$ of a default-free region, at which the value functions are differentiable, then*

$$(\rho-\lambda)\, u'(c_j) = \sum_{k\neq j} \eta_{jk}\,\big(u'(c_k) - u'(c_j)\big) \qquad \text{at } B^\circ. \tag{22}$$

In particular the state with the lowest consumption at B° cannot have an interior price-one steady state, so under the finite-reach condition, if the lowest state has a steady state at $\underline{B}$ then $\underline{B} = B_{\mathrm{pd}}(z_L, 1)$.

Condition (22) is the Euler equation at a stationary point: the impatience gain from borrowing a little balances the precautionary loss from arriving in a worse state with more debt. In the lowest state only the impatience term is present, so the sovereign borrows until it reaches its own default threshold, which is the steady state of that state. In a better state the two terms compete,

and there is a second force that the Euler equation does not show: beyond the threshold of a state that can be reached in one switch, a switch triggers an immediate first default, so the price schedule of the better state has a kink there and the marginal revenue of issuance drops. With two states this kink pins the high state to the low state's threshold at the calibrated default costs for every shock size and switching rate we tried: both states share a point mass at that debt level, or the high state stops below it, and default never occurs in the ergodic set. Default in the ergodic set requires a switch that lowers the threshold below the current debt, which happens when a state can jump to a state worse than the one its precautionary motive guards against: a rare direct jump to the bottom, or the continuum of the diffusion case, where the threshold moves continuously and the sovereign is caught on the wrong side of it.

Numerical illustration. Figure 4 shows the equilibrium with three states, $z \in \{-0.08, 0, +0.08\}$, at the calibrated parameters. Adjacent states switch at the rate $\eta = 0.096$, the autocorrelation decay rate of the calibrated diffusion ($\eta = \mu/2$ for a symmetric two-state chain, with $\mu = -\ln \rho_z$ the mean-reversion rate of Section 4.1), and the two upper states jump directly to the lowest one at the rate $\epsilon = 0.03$, once every three decades. The case is an illustration of the mechanics, not a quantitative exercise: the shock is smaller than the stationary standard deviation of the calibrated process, and the rare jump is the device that lets a chain catch the sovereign on the wrong side of a threshold. The solution is computed on a grid with 201 nodes on $[0, 1]$ with the standard at $\underline{q} = 0.7$, by the time stepping of Appendix B.2, to residuals of 10^{-7} and 2×10^{-9}. The default thresholds are $B = 0.325$, 0.440 and 0.565 in the low, middle and high state, and the default intensity jumps to 0.46 at the first. The middle state borrows up to the node just below its own threshold, $B = 0.435$, and stops there: the rare jump is too unlikely to hold it back, and it is pinned by its own threshold, as in the deterministic case. The high state stops earlier, at an interior steady state $B \approx 0.29$ where (22) balances its impatience against the lower consumption of the states it can switch to. The low state repays down to its own threshold, $B = 0.320$, from wherever it finds itself. The ergodic distribution, shown by state in the last panel, has atoms at the two thresholds, mass 0.27 at the middle state's and 0.16 at the low state's, a hump at the high state's steady state, and a continuous part on the low state's repayment path between the two thresholds. That path is where default happens: a jump from the middle state's atom lands the sovereign in partial default at $B = 0.435$, above the low state's threshold, and it withholds a share of its debt service while it repays. Default occurs on 0.07 of the ergodic mass, all of it in the low state, and mean debt is 0.37. Debt does not rise during these episodes: in the low state the sovereign repays from the moment it defaults, and borrowing into default requires the continuum of thresholds of the diffusion case, Section 4.3.

Existence and the price cliffs. Given policies, the prices are the Feynman–Kac values (11) and are continuous except at stationary points of debt that are not approached from both sides, where the paths from the two sides separate before the next switch. Without the underwriting standard

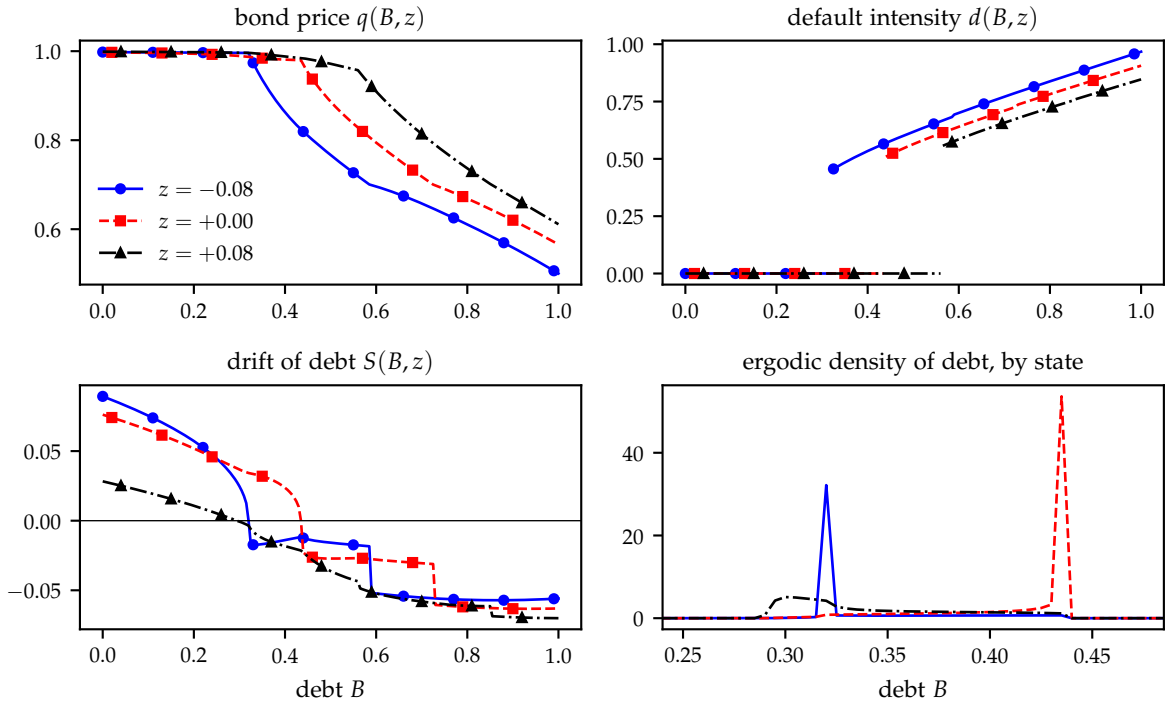


Figure 4: The three-state case: bond price, default intensity and drift of debt by productivity state, and the ergodic density of debt by state (atoms are drawn as spikes). Calibrated parameters; $z \in \{-0.08, 0, 0.08\}$, adjacent switching rate 0.096, rate of the direct jump to the lowest state 0.03.

the chain model has such a point in the lowest state at the full-default threshold, and the price jumps there; with it, every stationary point is approached from both sides and the prices are continuous. A general existence theorem for the coupled equilibrium is not available: the best-response map from prices to prices is discontinuous (thresholds move with q) and not monotone (a higher price lowers default today but raises borrowing), so neither Schauder's nor Tarski's theorem applies directly. We show in Appendix A that the best response does preserve the class of prices nonincreasing in debt.

4 Quantitative Analysis

This section establishes that the continuous-time model is the model of Arellano, Mateos-Planas, and Ríos-Rull (2023) quantitatively, by calibrating the diffusion case to their targets, and then shows the paper's claims at work in the calibrated model: the regions of default and the debt limit in the policies, arrears as borrowing in the shape of the ergodic set and in the response to a shock, and the standard at the edge of the state space.

4.1 The Diffusion Case

Log productivity follows an Ornstein–Uhlenbeck process, $dz_t = -\mu z_t\,dt + \sigma\,dW_t$, reflected at $\pm\bar{z}$. Its generator is $\mathcal{L}_z = -\mu z\,\partial_z + \frac{1}{2}\sigma^2\partial_{zz}$, so the HJB and pricing equations are

$$\rho V = \max_{c,d}\ \Big\{u(c) + S\,V_B - \mu z\,V_z + \tfrac{1}{2}\sigma^2 V_{zz}\Big\}, \tag{23}$$

$$\xi(d)\,q = (1-d)(\lambda+\delta) + \tilde{S}\,q_B - \mu z\,q_z + \tfrac{1}{2}\sigma^2 q_{zz}, \tag{24}$$

with Neumann conditions $V_z = q_z = 0$ at $\pm\bar{z}$, and the stationary density solves

$$0 = -\partial_B\big[\tilde{S}\,g\big] + \partial_z\big[\mu z\,g\big] + \tfrac{1}{2}\sigma^2\,\partial_{zz}g, \tag{25}$$

with zero-flux conditions $\mu z g + \frac{1}{2}\sigma^2 g_z = 0$ at $z = \pm\bar{z}$ and $\tilde{S}g = 0$ at $B = \bar{B}$, and an atom at $B = 0$ that collects the inflow where $\tilde{S}(0,z) < 0$. The process is the continuous-time counterpart of the annual AR(1) of Arellano, Mateos-Planas, and Ríos-Rull (2023): an AR(1) with persistence ρ_z and innovation standard deviation σ_ϵ observed at annual frequency corresponds to $\mu = -\ln\rho_z$ and a stationary standard deviation $\sigma_\epsilon/\sqrt{1-\rho_z^2} = \sigma/\sqrt{2\mu}$. Given a continuous price, (23) is a degenerate parabolic equation with a state constraint in B, whose value function is its unique constrained viscosity solution and is Lipschitz in B (Soner 1986; Capuzzo-Dolcetta and Lions 1990; Fleming and Soner 2006), and the upwind scheme of Appendix B.3 converges to it (Barles and Souganidis 1991). The coupled equilibrium is computed as the fixed point of the two equations.

4.2 Calibration and Computation

Strategy. Utility is CRRA, $u(c) = c^{1-\nu}/(1-\nu)$, and the default cost is (4), the level form of the fixed cost used by Arellano, Mateos-Planas, and Ríos-Rull (2023). We follow their calibration as closely as the continuous-time formulation allows. Risk aversion $\nu = 2$ and the risk-free rate $r = \ln 1.04$ are fixed. The nine parameters they estimate are our free parameters as well: the default cost parameters $\gamma_0, \gamma_1, \gamma_2, z^*$, the discount rate ρ (their $\beta = e^{-\rho}$), the recovery κ, the maturity rate δ (their survival rate is $1-\delta$), and the productivity parameters ρ_z, σ_ϵ mapped as above. The targets are the eleven moments of their Table 4, with their weights (three on the mean of debt due to output, 1.25 on the standard deviation of the spread, one on the rest), and the loss is the weighted sum of squared proportional deviations. The underwriting standard is set at $\underline{q} = 0.7$ (Section 3.2).

Accounting and moments. Debt due is $(\delta+\lambda)B$ per year, debt service is $(1-d)$ times debt due, defaulted coupons are d times debt due, and the annual partial default is defaulted coupons over debt due within the year, as in their equation (4). The debt level is B (face value, risk-free price one). Output and consumption are annual integrals, logged and linearly detrended. A year is in partial default if any coupon was defaulted on during it; at the calibrated point the instantaneous probability of being in default under the stationary distribution is 0.28, against

an annual frequency of 0.38, the difference being time aggregation. The untargeted moments conditional on default are computed on the simulated annual paths as well. Moments that are functions of the state (debt, debt service and debt due to output, the spread and its correlations) are computed exactly from the stationary distribution of the discretized process, by solving the adjoint system of Appendix B.3. Moments that are path functionals under annual accounting (the partial default frequency, mean and standard deviation, the persistence and volatility of output, consumption) are computed from sixteen simulated paths of 25,000 years each.

Parameters. Table 1 reports the parameters next to those of Arellano, Mateos-Planas, and Ríos-Rull (2023). We calibrated the model twice: first without the underwriting standard, which gave the parameters at which Section 3.2 and Appendix C document the behavior at high debt, then with the standard in place, which gives Table 1 and everything else in the paper. The re-calibration under the standard stopped after 285 evaluations. The calibrated values are close to theirs: the default cost is somewhat lower and more convex, the discount rate and the maturity rate somewhat higher, and the productivity process is essentially theirs.

Table 1: Calibration

		AMR (2023)	This paper
Fixed			
ν	risk aversion	2	2
r	risk-free rate ($\ln 1.04$)	0.039	0.039
λ	coupon, $= r$	0.039	0.039
q	underwriting standard	–	0.70
Calibrated			
γ_0	variable default cost, level	0.0476	0.0587
γ_1	variable default cost, curvature	2.00	3.22
γ_2	fixed default cost, slope	0.120	0.057
z^*	fixed default cost, threshold (level)	0.938	0.745
ρ	discount rate ($\beta = e^{-\rho}$)	0.047	0.058
κ	recovery of defaulted coupons	0.70	0.56
δ	maturity rate (survival $1-\delta$)	0.12	0.17
ρ_z	annual persistence of productivity	0.875	0.826
σ_ϵ	annual innovation s.d.	0.052	0.063

Notes: The discrete-time parameters of Arellano, Mateos-Planas, and Ríos-Rull (2023) (Table 3) are mapped to continuous time by $\rho = -\ln\beta$, $\delta = 1 -$ survival rate, $\mu = -\ln\rho_z$, $\sigma = \sigma_\epsilon\sqrt{2\mu/(1-\rho_z^2)}$. The free parameters are chosen to minimize the weighted distance to the targets of Table 2, as in their calibration.

Computation. The equilibrium is computed by the implicit upwind finite difference scheme of Achdou et al. (2022), with the value and the price updated jointly at each implicit step because the alternating iteration between them is unstable, on a grid of 201 nodes in debt on $[0, 1]$ and 75 nodes in productivity on ± 2.5 stationary standard deviations. Each parameter evaluation

in the calibration solves the model from a warm start in about two minutes.[5] At the reported point the HJB residual is 5×10^{-3} and the pricing residual 1×10^{-2}; the latter is carried by the nodes at the full-default cliff, where the first-order scheme cannot determine the sign of the drift (Section 3.2), and these nodes hold no ergodic mass. No taste shocks or other smoothing of the decision problem are needed: the default intensity is a continuous control, the only non-concavity is the fixed cost's jump at $d = 0$, and the monotone three-candidate scheme handles it directly. Discrete-time default models with long-term debt rely on i.i.d. shocks to the default and borrowing choices for the price schedule to converge (Chatterjee and Eyigungor 2012; Mihalache 2025); the devices are effective and widely used, and the continuous formulation simply does not need them, which removes both the conceptual distance between the smoothed and the actual decision and the dependence of convergence on the scale of the shocks. Appendix B describes the scheme, the treatment of the non-concave Hamiltonian created by the fixed cost, the initial guess, the convergence criteria, and the exact marching solution of the deterministic case that we use as a check. Section 3.2 describes the difficulties the model poses at high debt and how the underwriting standard resolves them.

4.3 Quantitative Properties

Fit. Table 2 reports the targeted moments in the data, in the model of Arellano, Mateos-Planas, and Ríos-Rull (2023) and in ours, and the untargeted moments of their Table 4. The loss at the current point is 0.237 against 0.228 for their model under the same weights (most of which comes from their standard deviation of debt to output). The continuous-time model reproduces the frequency of partial default, the mean and dispersion of the default intensity in default, the debt service and debt due ratios and the output process; it carries somewhat less debt than the data and generates a spread volatility below the target. Among the untargeted moments, defaulted coupons and the relative volatility of consumption are close to the data; the spread is more strongly correlated with output and with debt than in the data, as in their model.

Values, policies and prices. Figure 5 shows the value, the bond price, consumption, the default intensity and the drift of debt as functions of debt at four productivity levels, and the ergodic marginal distribution of debt; Figure 6 shows the price, the default intensity and the ergodic density as contour maps over the state space, with the zero-drift locus. The value is decreasing in debt and increasing in productivity. The default policy has the structure of Proposition 4. At the lowest productivity shown, $z = -0.28$, which lies just above the fixed-cost threshold $\ln z^* = -0.29$ (at the calibrated parameters the free-default region sits below the lowest node of the productivity grid and is visited less than half a percent of the time), the no-default region is a sliver at zero debt and the sovereign defaults at almost any positive debt, with an intensity that rises with debt to full default; at average and high productivity it does not default until debt

5. The numerical calibration itself was run with 25 productivity nodes, for expediency; every result reported in the paper is computed at the calibrated parameters on the 75-node grid.

Table 2: Moments in Data and Model

Moment	Weight	Data	AMR (2023)	This paper
Targeted				
Partial default: frequency	1	0.36	0.37	0.38
Partial default: mean $\mid d > 0$	1	0.38	0.39	0.42
Partial default: s.d. $\mid d > 0$	1	0.22	0.19	0.22
Debt to output: mean	1	0.32	0.32	0.24
Debt to output: s.d.	1	0.18	0.25	0.17
Debt service to output: mean	1	0.036	0.035	0.034
Debt service to output: s.d.	1	0.021	0.022	0.020
Debt due to output: mean	3	0.049	0.050	0.049
Spread: s.d.	1.25	0.041	0.037	0.027
Output: persistence	1	0.89	0.88	0.87
Output: s.d.	1	0.10	0.12	0.10
Loss			0.228	0.237
Untargeted				
Defaulted coupons to output: mean $\mid d > 0$		0.052	0.040	0.040
Defaulted coupons to output: s.d. $\mid d > 0$		0.064	0.036	0.025
Spread: mean		0.053	0.016	0.024
Spread: correlation with output		-0.17	-0.38	-0.80
Spread: correlation with debt		0.24	0.56	0.87
Consumption s.d. relative to output s.d.		1.00	0.91	0.93

Notes: Data and the discrete-time model from Arellano, Mateos-Planas, and Ríos-Rull (2023), Table 4. The loss is the weighted sum of squared proportional deviations from the data.

reaches the threshold $B_{\text{pd}}(z, q)$, where the default intensity jumps to the first default intensity and rises thereafter. The threshold rises with productivity, so the no-default region is a wedge in the (B, z) plane, and the full-default line is reached at lower debt in worse times. The price is one in the no-default region, falls at the default threshold, and declines toward the floor in the default region; its contours in Figure 6 track the default thresholds. Consumption drops at the default threshold by the fixed cost and the lost issuance revenue, and is nearly flat in debt in the default region, where the sovereign smooths by withholding payments rather than by issuing.

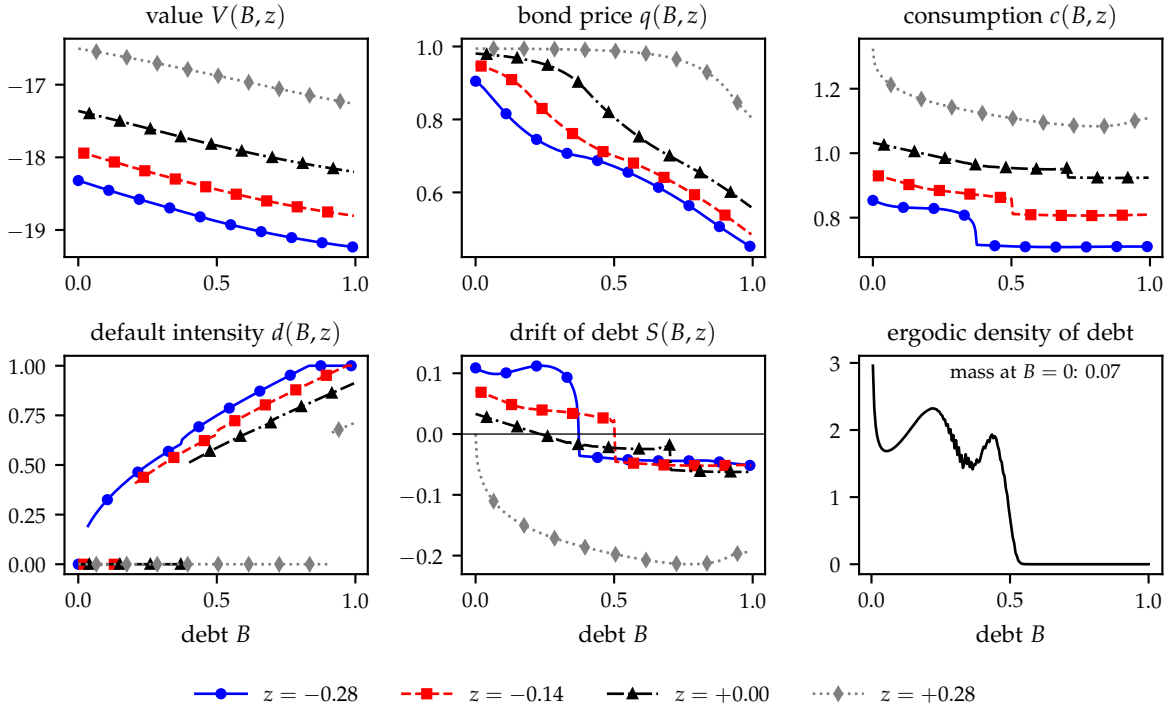


Figure 5: The diffusion case: value, bond price, consumption, default intensity and drift of debt as functions of debt at four productivity levels, and the ergodic marginal density of debt (the atom at $B = 0$ is reported in the panel).

The drift and the ergodic set. The drift of debt is positive at low debt and negative at high debt in every productivity state, with a single zero-drift locus that is downward sloping: the sovereign borrows up to a target debt level that is higher in bad times, because the no-default threshold is lower there and the sovereign uses partial default as a source of financing, and lower in good times, when the precautionary motive dominates and the sovereign repays. This is arrears as borrowing in the calibrated model: the sovereign's debt target rises when the market price falls, because the price of the arrears channel does not. At the highest productivity the drift is negative at every debt level and the sovereign repays all the way to zero debt, which is the atom of the ergodic distribution at $B = 0$. The ergodic set is the region swept by the zero-drift locus as productivity moves: the density in Figure 6 concentrates along the locus, with most of the mass between zero debt and the no-default threshold at average productivity, $B_{\text{pd}}(0, 1) = 0.456$, and

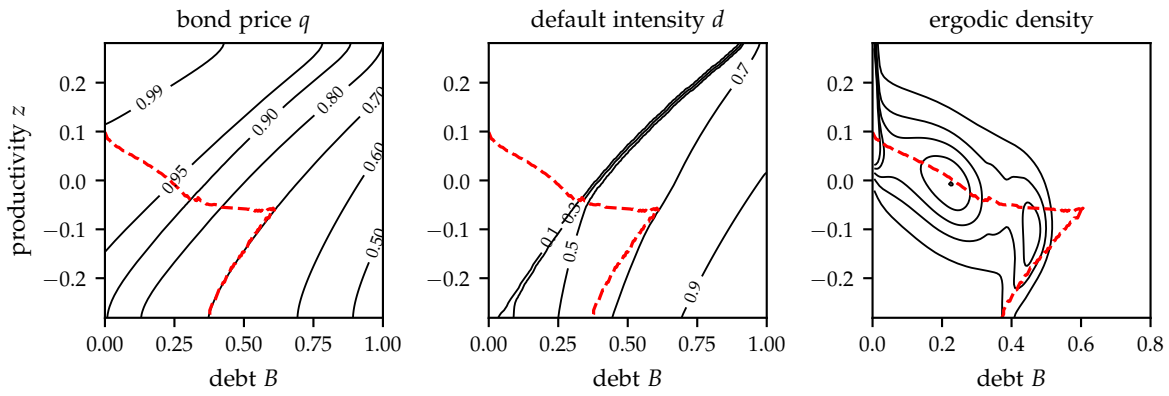


Figure 6: The diffusion case over the state space: contours of the bond price, of the default intensity, and of the ergodic density. Dashed: the zero-drift locus, where debt is stationary.

a tail that extends into the partial-default region at low productivity. It ends well before the top of the grid: the standard guarantees that the full-default region is transient, and the sovereign's own choice keeps it inward of $B \approx 0.6$. Spreads are therefore low most of the time and high in the tail, which is where the model's spread volatility comes from.

Impulse responses. Figure 7 shows the response of the economy to a negative productivity shock. We start from the ergodic distribution, lower z by 0.10 (about one stationary standard deviation, a ten percent fall in output) at date zero, and report the expected path of each variable minus its ergodic mean. The expectation is computed exactly, by evolving the shifted distribution with the forward equation of the discretized process, so no simulation is involved (Appendix B.3). Output falls on impact and recovers at the rate μ. Consumption falls by less than output and recovers more slowly: the sovereign smooths the shock by withholding debt service, so the default intensity jumps on impact and stays elevated for a decade. The withheld payments accrue as arrears and debt *rises* for about six years after the shock, by some six percent of annual output at the peak, before it is worked off; the spread rises on impact by about 140 basis points and declines with productivity. The last panel decomposes the financing of the shock (new issues in face value and defaulted coupons, both in percent of mean output): on impact the sovereign issues new debt, but as debt rises and the price falls, defaulted coupons take over as the main source of resources, and new issuance becomes negligible while the default intensity is still high. These are the dynamics of partial default episodes documented by Arellano, Mateos-Planas, and Ríos-Rull (2023): default is protracted, debt rises during the episode, and the economy exits the episode with more debt than it entered it, as productivity recovers. In the model they are the direct expression of the first claim: the sovereign finances the shock through arrears, at a price that the market does not set, and stays in the market while it does.

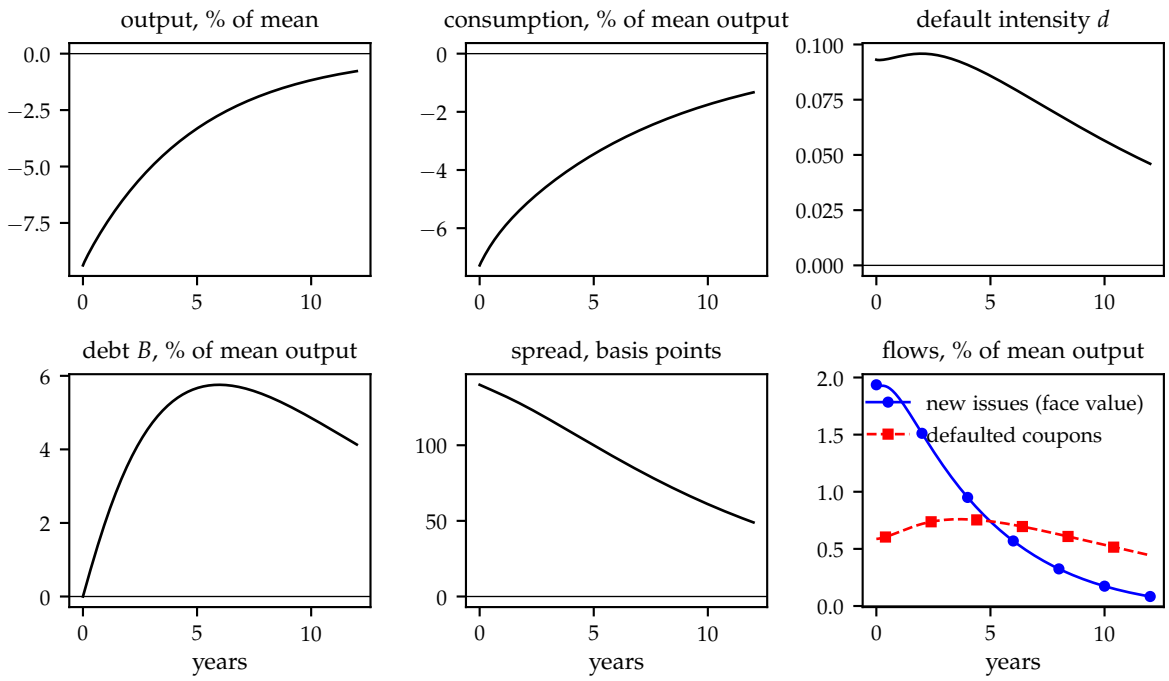


Figure 7: Impulse responses to a negative productivity shock of 0.10 at date zero, from the ergodic distribution: expected path under the forward equation minus the ergodic mean. Flows are in percent of mean annual output; the spread in basis points.

5 Conclusion

Arellano, Mateos-Planas, and Ríos-Rull (2023) replaced the binary default of the standard sovereign debt model with a continuous one and showed that the resulting model matches what default episodes look like. We have studied their model in continuous time, where it is tractable, and asked what partial default does. Three answers organize the paper. Arrears are borrowing: withholding debt service issues debt to existing creditors at an implicit price that does not depend on the market, paid for with the output cost of default, which is why the default decision is a static financing decision, why sovereigns never default a little, why the debt a sovereign will service in full is a closed-form limit with a soft edge, and why debt targets rise in bad times and debt rises through arrears after a bad shock while the sovereign stays in the market. The model has a genuine pathology at high debt, an explosion of debt where default is cheap whose entry point looks like a second steady state on a grid, and it is a property of the equilibrium, not of any method. An underwriting standard on lenders is the minimal repair: it makes the equilibrium well posed and the solution independent of the grid, and it makes the sovereign default more where it binds. Calibrated to the same targets, the continuous-time model fits them comparably, reproduces the protracted, debt-increasing episodes that motivate the theory, and does so fast, exactly, and without smoothing devices.

Several avenues remain. On the theoretical side, the multiplicity question of Section 3.3 is sharp and open: whether self-fulfilling partial-default steady states exist depends, we conjecture,

on the impatience wedge relative to the spread that the first default implies, and a proof would give a continuous-time account of slow-moving debt crises with arrears. Existence of the coupled equilibrium in the stochastic cases is open for reasons we have made explicit. On the quantitative side, the identity behind our first claim suggests where to look next: the implicit price of arrears is set by the recovery parameter and does not respond to the market, so any force that moves the market price without moving the recovery, lenders' pricing kernels, their underwriting standards, the maturity of the debt, changes how much of a sovereign's financing comes through arrears, and the data on arrears could discipline it. The natural extensions carry larger state spaces: a measure of small open economies exposed to common global shocks, shocks to the lenders' side, reserves, or a maturity composition. A tensor grid becomes expensive with a third state and infeasible with a fourth. The neural network methods designed for this case do not, in our experiment of Appendix D, recover the finite difference solution even in two dimensions, because the stationary residual they minimize does not select the right solution; a causal, time-stepping version of these methods, or a reinforcement learning formulation that respects the direction of time, is the route we see for larger state spaces.

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Appendix to "Arrears as Borrowing: The Economics of Partial Default"

by Sangdong Kim and Gabriel Mihalache

Appendix A Proofs and Derivations

Subsection A.1 Derivations

Derivation of the HJB equation (8). Over a short interval Δt the principle of optimality gives $V(B_0, z_0) = \max\{u(c_0)\Delta t + e^{-\rho\Delta t}\mathbb{E}_0 V(B_{\Delta t}, z_{\Delta t})\}$. Expanding, $\mathbb{E}_0 V(B_{\Delta t}, z_{\Delta t}) = V + V_B\, S\, \Delta t + \mathcal{L}_z V\, \Delta t + o(\Delta t)$, where $\mathcal{L}_z$ is the generator of z (zero for constant z, $\sum_{k\neq j}\eta_{jk}[V(B, z_k) - V(B, z_j)]$ for a chain, $-\mu z V_z + \frac{1}{2}\sigma^2 V_{zz}$ for the diffusion). Dividing by Δt and letting $\Delta t \to 0$ gives (8).

Derivation of the pricing equation (12). Over $[t, t+\Delta t]$ a unit of debt pays $(1-d_t)(\delta+\lambda)\Delta t$, matures at rate δ, grows by the arrears $\kappa(\delta+\lambda)d_t\Delta t$, and is discounted at r, so $q_t = (1-d_t)(\delta+\lambda)\Delta t + e^{-\xi(d_t)\Delta t}\,\mathbb{E}_t q_{t+\Delta t} + o(\Delta t)$ with $\xi(d) = r + \delta - \kappa(\delta+\lambda)d$. The same expansion gives (12); integrating gives (11).

Derivation of the forward equation (14). For a test function φ, $\frac{d}{dt}\mathbb{E}\varphi(B_t, z_t) = \mathbb{E}[\tilde{S}\varphi_B + \mathcal{L}_z\varphi]$, and writing the expectation against the density g and integrating by parts gives $\partial_t g = -\partial_B[\tilde{S}g] + \mathcal{L}_z^* g$, with $\mathcal{L}_z^* g = \partial_z[\mu z g] + \frac{1}{2}\sigma^2\partial_{zz} g$ in the diffusion case. The discrete analogue is exact: if A is the generator matrix of the discretized process, the stationary distribution solves $A^\top g = 0$.

Subsection A.2 Proofs for Section 3.1

Proof of Proposition 2. With $\ell = n - \kappa(\delta+\lambda)dB$, (1) gives $c = \phi e^z - (1-d)(\delta+\lambda)B + qn - \kappa q(\delta+\lambda)dB = \phi e^z - (\delta+\lambda)B + qn + (1-\kappa q)(\delta+\lambda)Bd$. At a given drift S, $n = S + \delta B$, so $c = \mathcal{R}(d) + qS$ with $\mathcal{R}$ as stated, and $\mathcal{R}_d$ follows from (4). □

Proof of Theorem 3. Fix any admissible drift S. The objective $u(\mathcal{R}(d) + qS) + SV_B$ depends on d only through $\mathcal{R}(d)$, u is increasing, and the feasibility constraint $c \geq c_{\min}$ is relaxed by a larger $\mathcal{R}(d)$. Hence for every S the best d maximizes $\mathcal{R}$, and so does the d-component of the joint maximizer. □

Proof of Proposition 4. On $(0, 1]$, $\mathcal{R}'(d) = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1} + (1-\kappa q)(\delta+\lambda)B$ is strictly decreasing with $\mathcal{R}'(0^+) > 0$, so $\mathcal{R}$ is strictly concave with maximizer d_{int} of (10). The gain from defaulting is $\mathcal{R}(d_{\text{int}}) - \mathcal{R}(0) = \Delta(B, z, q) - (1 - F(z))e^z$, which gives (16). For an interior d_{int}, $\Delta = (1 - 1/\gamma_1)(1-\kappa q)(\delta+\lambda)Bd_{\text{int}} \propto B^{\gamma_1/(\gamma_1-1)}$, which is increasing in B, decreasing in q and zero at $B = 0$; (a), (b) and (d) follow. Eliminating B between $\Delta = (1-F)e^z$ and (10) gives (17), and eliminating d_{int} gives (18). □

Proof of Proposition 6. (11) is the Feynman–Kac representation of (12) with discount rate $\xi(d_t) \geq \underline{\xi}(1) > 0$. The bound: $(1-d)(\lambda+\delta) \leq \bar{q}\,\xi(d)$ for all $d \in [0,1]$ is equivalent to $d\,\xi(1) \geq 0$; integrating, $q \leq \bar{q}\,\mathbb{E}\int_0^\infty \xi(d_t)e^{-\int_0^t \xi}dt = \bar{q}$, with equality iff $d \equiv 0$. □

Proof of Proposition 7. With the drift bounded away from zero, from $(B_0+\eta, z_0)$ debt reaches B_0 at a time $\tau(\eta) \to 0$ on every path along which productivity stays in the neighborhood. By (11) and the strong Markov property, $q(B_0+\eta, z_0) = \mathbb{E}\big[\int_0^\tau e^{-\int_0^t \xi}(1-d)(\lambda+\delta)\,dt + e^{-\int_0^\tau \xi}\,q(B_\tau, z_\tau)\big]$. The integral is $O(\tau)$ and the discount factor tends to one. In the deterministic model $(B_\tau, z_\tau) = (B_0, z_0)$. For a chain, productivity switches before τ with probability $O(\tau)$, and on the complement $(B_\tau, z_\tau) = (B_0, z_0)$; since q is bounded, the expectation tends to $q(B_0, z_0)$. For the diffusion, productivity leaves the neighborhood before τ with vanishing probability and otherwise $B_\tau = B_0$ with $z_\tau - z_0 = O_p(\tau^{1/2})$, so the expectation tends to $q(B_0, z_0)$ when $q(B_0, \cdot)$ is continuous at z_0. □

Proof of Remark 8. The default rule under $q = 0$ is Theorem 3 with $q = 0$. With $q = 0$ issuance does not enter the budget, so any ℓ is optimal; with $d = 1$ no payment is made and $q = 0$ is the lenders' valuation; invariance is needed for self-consistency, and with $d = 1$ debt decays at rate a unless $\ell \geq aB$. For the removal: suppose $q(B_0, z_0) = 0$. By Proposition 6, $d = 1$ a.e. along the path, so $q = 0$ along the path, so $\ell = 0$ along the path and $B_t = B_0 e^{-at}$. Since $B_{\mathrm{fd}}(z, 0) = \gamma_0\gamma_1 F(z)e^z/(\delta+\lambda)$ is bounded below by a positive constant on the bounded productivity ranges of Sections 3.4 and 4.1, in finite time $B_t < \inf_z B_{\mathrm{fd}}(z, 0)$ and the sovereign defaults partially, a contradiction. □

Proof of Proposition 9. Let $B' < B$ and let (d_t, ℓ_t) be the realized default and issuance path of the optimal policy from (B, z) along the realized z-path, with debt B_t. The sovereign at B' runs the same default path and the issuance $\ell'_t = \ell_t$ if $\ell_t \geq 0$, $\ell'_t = \ell_t\, q(B_t, z_t)/q(B'_t, z_t)$ if $\ell_t < 0$ (the same bonds when issuing, the same expenditure when buying back), with $\ell'_t = 0$ if $B'_t = 0$ and $\ell_t < 0$. The gap $g_t = B_t - B'_t$ solves $\dot{g} = (\ell_t - \ell'_t) - a(d_t)g$. While $g \geq 0$, $q(B'_t, z_t) \geq q(B_t, z_t)$, hence $\ell_t - \ell'_t \leq 0$ and g is nonincreasing. If g were negative on some interval $(t_2, t_1]$ with $g(t_2) = 0$, then on it $\ell_t - \ell'_t \geq 0$ and $-a(d_t)g > 0$, so $\dot{g} > 0$ and $g(t_1) > 0$, a contradiction. Hence $0 \leq B'_t \leq B_t$: the state constraints hold, and the standard is respected because $q(B'_t, z_t) < \underline{q}$ implies $q(B_t, z_t) < \underline{q}$, hence $\ell_t \leq 0$ and $\ell'_t \leq 0$. Consumption satisfies $c'_t - c_t = (1-d_t)(\delta+\lambda)g_t + [q(B'_t, z_t)\ell'_t - q(B_t, z_t)\ell_t] \geq 0$, the bracket being 0 when $\ell_t < 0$, $\ell_t[q(B'_t, z_t) - q(B_t, z_t)] \geq 0$ when $\ell_t \geq 0$, and $-q(B_t, z_t)\ell_t > 0$ when the override is active ($B'_t = 0$ and $\ell_t < 0$). The default costs coincide, so the mimicking path gives the sovereign at B' at least the utility of the optimal path from B. □

Subsection A.3 Proofs for Section 3.2

Proof of Proposition 10. Parametrize the choice by (d, ℓ). The objective is $G(d, \ell) = u(\tilde{\mathcal{R}}(d) + q\ell) + (\ell - a(d)B)V_B$ with $\tilde{\mathcal{R}}(d) = \phi(d, z)e^z - (1-d)(\delta+\lambda)B$ the resource function at a zero price, and the constraint is $\ell \leq 0$, which no longer involves d. On $(0, 1] \times \mathbb{R}$, $\tilde{\mathcal{R}}$ is concave in d, u is concave

and increasing and the last term is linear, so G is jointly concave. (a) is immediate. For (b), let (d_c, ℓ_c) be a constrained maximizer on the $d > 0$ branch with $\ell_c < 0$; the segment from it to (d^*, ℓ^*) crosses $\{\ell = 0\}$ at a point P, and by concavity $G(P) \geq \min\{G(d_c, \ell_c), G(d^*, \ell^*)\} = G(d_c, \ell_c)$, so P is a constrained maximizer with $\ell = 0$ on the $d > 0$ branch. The $d = 0$ branch is treated separately, since G is not concave across the fixed-cost jump at $d = 0$: its own unconstrained maximizer has issuance $\ell_0^* > \ell^* > 0$, because $d^* > 0$ maximizes $\mathcal{R}$ and $a(d^*) < \delta$, so the same comparison along $\{d = 0\}$ gives $\ell = 0$ there as well, and the constrained optimum is the better of $J(d^{**})$ and $J(0)$. On $\{\ell = 0\}$, $c = \tilde{\mathcal{R}}(d)$ and $S = -a(d)B$, which gives J. At an interior maximizer $J'(d) = u'(c)\tilde{\mathcal{R}}'(d) + \kappa(\delta + \lambda)BV_B = 0$, that is $\tilde{\mathcal{R}}'(d) = \kappa(\delta + \lambda)B\hat{q}$, the first-order condition of $\mathcal{R}(\cdot; B, z, \hat{q})$. Finally $\partial_\ell G(d^{**}, 0) = qu'(c^{**}) + V_B > 0$ because the constraint binds strictly, which is $\hat{q} < q$, and $\hat{q} > 0$ because $V_B < 0$ (Proposition 9). □

Proof of Proposition 11. (a) With $d = 1$ and $\ell \leq 0$, $S = \ell - aB \leq -aB$; integrate. (b) In $\{q < \underline{q}\}$ the drift is $S \leq -a(d)B \leq -aB < 0$ for every d, so there is no zero there. At a stationary point on the floor locus the paths from the two sides both reach the point; if they do so in finite time, the Feynman–Kac value (11) from either side converges to the value at the point, as in the proof of Theorem 14. (c) At $\bar{B}$ the sovereign is in full default, hence $S < 0$ there in every state, and a state with inward drift carries no stationary mass. (d) The restriction binds only where $q < \underline{q} < 1$, which excludes the no-default region. □

Proof of Proposition 12. Debt rises only when $\ell > 0$, which the standard permits only at $q \geq \underline{q}$, hence only at $B \leq B^\sharp$; above $B^\sharp$, $\ell \leq 0$ and $S = \ell - a(d)B \leq -aB$. The rest follows from the continuity of B_t. □

Subsection A.4 Proofs for Section 3.3

Proof of Theorem 14. From any $B \leq B^*$ the path rises to B^* and stays, with $d \equiv 0$; (11) gives $q = 1$ and consumption at B^* is $1 - (\delta + \lambda)B^* + \delta B^* = 1 - \lambda B^*$ forever. At B^* the price is one and $d = 0$ is optimal, so $B^* \leq B_{\mathrm{pd}}(0, 1)$. Suppose $B^* < B_{\mathrm{pd}}(0, 1)$. By finite reach and (11), $q(B^* + \eta) \geq e^{-(r+\delta)\tau(\eta)} \to 1$, and $B_{\mathrm{pd}}(0, \cdot)$ is continuous and increasing, so for small η, $B^* + \eta < B_{\mathrm{pd}}(0, q(B^* + \eta))$ and $d = 0$ at $B^* + \eta$; along the path down to B^* then $d \equiv 0$ and $q(B^* + \eta) = 1$ exactly. Thus $q \equiv 1$ and $d \equiv 0$ on a neighborhood of B^*, where the sovereign faces a constant price of one with no default, and staying at B^* is not optimal because $\rho > \lambda$: the round trip that borrows at rate $s = \epsilon\tau$ for a time τ and repays at the same rate for the next τ perturbs consumption by $\Delta c_t = s - \lambda s t$ on $[0, \tau]$ and $-s - \lambda s(2\tau - t)$ on $[\tau, 2\tau]$, with first-order gain $u'(\bar{c}) \int_0^{2\tau} e^{-\rho t} \Delta c_t \, dt = (\rho - \lambda)u'(\bar{c})\epsilon\tau^3 + O(\epsilon\tau^4)$ and second-order loss $O(\epsilon^2\tau^3)$; for small ϵ the gain dominates, contradicting $S(B^*) = 0$. □

Proof of Theorem 16. Step 1: the drift vanishes. On the right branch the first-order condition is $V' = -qu'(c)$, and with $S = (c - \mathcal{R})/q$, $\mathcal{R}(B) := \max_d \mathcal{R}(d; B, 0, q(B))$, the HJB reads

$$\rho V = u(c) + w\, u'(c), \qquad w := \mathcal{R} - c = -qS \geq 0. \tag{26}$$

Let $x \downarrow 0$: $V \to u(\bar{c})/\rho$, $q \to 1$ (by the finite-reach condition and (11), as in the proof of Theorem 14) and $\mathcal{R} \to \bar{c}$, because at $B^* = B_{\mathrm{pd}}(0,1)$ the sovereign is indifferent between $d = 0$ and d_{first}. Along any sequence on which $c \to c^+$, (26) gives $u(\bar{c}) = u(c^+) - (c^+ - \bar{c})u'(c^+)$; the function $g(c) = u(c) - (c - \bar{c})u'(c) - u(\bar{c})$ has $g(\bar{c}) = 0$ and $g'(c) = -u''(c)(c - \bar{c})$, so $g > 0$ for $c \neq \bar{c}$; hence $c \to \bar{c}$, $w \to 0$, $S \to 0$, and $V'(B^{*+}) = -u'(\bar{c})$, which is also the left derivative.

Step 2: the exponents. Put $p := 1 - q$. The pricing equation gives $p' = q\Pi/w$ with $\Pi(B,q) := \xi(d^*)q - (1-d^*)(\lambda+\delta) \to \Pi_0 > 0$. Differentiating (26), using $V' = -qu'(c)$, $c = \mathcal{R} - w$ and the envelope theorem $\mathcal{R}' = \mathcal{R}_B - \mathcal{R}_q p'$ with $\mathcal{R}_B = -(1-d^*)(\delta+\lambda) + qa(d^*)$ and $\mathcal{R}_q = a(d^*)B$,

$$w' = \mathcal{R}_B - \mathcal{R}_q p' + \frac{u'(c)}{|u''(c)|}\,\frac{\mathcal{R}_q q\Pi/w - (\mathcal{R}_B + \rho q)}{w} = \frac{K(x)}{w^2} + O(1/w),$$
$$K(x) \to K := \frac{u'(\bar{c})}{|u''(\bar{c})|}\,a(d_{\mathrm{first}})B^*\Pi_0.$$

Since $w \to 0$, $w^2 w' = K + o(1)$ and $w^3 = 3Kx(1+o(1))$, that is $w = \beta x^{1/3}(1+o(1))$ with $\beta^3 = 3K$. Then $p' = \Pi_0(1+o(1))/w$ integrates to $p = \alpha x^{2/3}(1+o(1))$, and $S = -w/q$, $c = \mathcal{R} - w$ give the stated laws; $V'' = -q'u'(c) - qu''(c)c'$ is dominated by $-|u''(\bar{c})|\beta x^{-2/3}/3$.

Step 3: the regular system (existence, uniqueness and analyticity of the branch). In the variable $s = x^{1/3}$ the scaled unknowns $Q = (1-q)/s^2$ and $\Sigma = -qS/s$ solve an analytic system of Briot–Bouquet type. With $p = s^2 Q$, $w = s\Sigma$ and $d/dx = (3s^2)^{-1} d/ds$,

$$sQ_s = \frac{3q\Pi}{\Sigma} - 2Q, \qquad s\Sigma_s = \frac{3u'}{|u''|}\,\frac{\mathcal{R}_q q\Pi}{\Sigma^2} - \Sigma - 3s\Big[\frac{u'}{|u''|}\,\frac{\mathcal{R}_B + \rho q}{\Sigma} + \frac{\mathcal{R}_q q\Pi}{\Sigma}\Big] + 3s^2\mathcal{R}_B,$$

with the coefficients evaluated at $c = \mathcal{R}(B^* + s^3, 1 - s^2 Q) - s\Sigma$ and $(B^* + s^3, 1 - s^2 Q)$. For CRRA utility and $d_{\mathrm{first}} \in (0,1)$ all coefficients are analytic near $(0, \alpha, \beta)$. At $s = 0$ the right-hand sides vanish exactly at (α, β) and their Jacobian is $\begin{pmatrix} -2 & -3\Pi_0/\beta^2 \\ 0 & -3 \end{pmatrix}$ (the $(2,2)$ entry is $-6(u'/|u''|)a(d_{\mathrm{first}})B^*\Pi_0/\beta^3 - 1 = -3$ by the definition of β). Neither eigenvalue is a positive integer, so the Briot–Bouquet theorem (Briot and Bouquet 1856; Hille 1976, Chapter 12) yields a unique solution analytic at $s = 0$ with the prescribed value. Since both eigenvalues are negative, every other solution of the linearized system behaves like s^{-2} or s^{-3} and is unbounded as $s \downarrow 0$, which is the sense in which the branch is unique.

Step 4: the second-order coefficients. The analytic solution has $Q = \alpha + Q_1 s + O(s^2)$ and $\Sigma = \beta + \Sigma_1 s + O(s^2)$. In the coefficients of the system, Π, $\mathcal{R}_q$, $\mathcal{R}_B$, q and d^* differ from their values Π_0, $a(d_{\mathrm{first}})B^*$, R_B, 1 and d_{first} at $(B^*, 1)$ by $O(s^2)$, while $\varphi(c) = u'(c)/|u''(c)|$ is evaluated at $c = \bar{c} - \beta s + O(s^2)$, so $\varphi = \varphi(\bar{c}) - \varphi'(\bar{c})\beta s + O(s^2)$. Collecting the terms of order s in the Σ equation, with $3\varphi(\bar{c})a(d_{\mathrm{first}})B^*\Pi_0 = \beta^3$, gives $\Sigma_1 = -\beta^2\varphi'(\bar{c})/\varphi(\bar{c}) - 3\Sigma_1 - 3\varphi(\bar{c})(R_B + \rho)/\beta - 3a(d_{\mathrm{first}})B^*\Pi_0/\beta$, which is the formula in the statement after $3a(d_{\mathrm{first}})B^*\Pi_0/\beta = 2a(d_{\mathrm{first}})B^*\alpha$; the terms of order s in the Q equation give $Q_1 = -3\Pi_0\Sigma_1/\beta^2 - 2Q_1$. Finally $-S = w/q = s\Sigma\,(1 + O(s^2))$ and $1 - q = s^2 Q$. □

Proof of Theorem 17. Away from B^* the system (19) is a regular ODE as long as $S^* < 0$, $c^* > c_{\min}$ and $d^* \in (0,1)$, since $(B,q) \mapsto d^*, \mathcal{R}^*$ are analytic there, so Picard–Lindelöf continues the branch uniquely from the cube-root start up to the first point where one of these fails or the standard starts to bind. On $\{q < \underline{q}\}$ with $\ell^* > 0$ the policy is the constrained one of Proposition 10(b), $S = -a(d^{**})B < 0$, and the HJB equation $\rho V = u(c^{**}) - a(d^{**})BV'$ together with the pricing equation is again a regular system in (V,q), continued in the same way up to the full-default threshold; at the switch consumption drops by $q\ell^*$ and V' may have a kink. Beyond the full-default threshold, with $\underline{q}$ above the full-default prices the sovereign cannot issue, and by hypothesis it does not buy back, so $S = -aB$, the pricing equation $\xi(1)q = -aBq'$ gives the power law, and $V(B) = \int_0^\tau e^{-\rho t} u\big((1-\gamma_0)F(0)\big)dt + e^{-\rho\tau} V(B_{\mathrm{fd}})$ with $\tau = a^{-1}\ln(B/B_{\mathrm{fd}})$. Continuity of q follows from Proposition 7, as the drift is negative throughout. □

Lemma 20 (local conditions at a partial-default steady state)**.** *Suppose the deterministic equilibrium has a steady state B° in the interior of the partial-default region at which S and q are differentiable, with $d^\circ = d^*(B^\circ, q^\circ) \in (0,1)$. Then $q^\circ = (1-d^\circ)(\lambda+\delta)/\xi(d^\circ)$, the slope of the price is pinned by the impatience wedge,*

$$-\frac{q'(B^\circ)}{q^\circ} = \frac{\rho - r}{a(d^\circ)B^\circ},$$

and the slope of the drift follows from the pricing equation,

$$q'(B^\circ)\big(S'(B^\circ) - \xi(d^\circ)\big) = (\lambda+\delta)(1-\kappa q^\circ)\,\frac{d}{dB} d_{\mathrm{int}}\big(B, q(B)\big)\Big|_{B^\circ},$$

so that the steady state is attracting, $S'(B^\circ) < 0$, if and only if the dilution term on the right exceeds $\xi(d^\circ)\,|q'(B^\circ)|$.

Proof of Lemma 20. The first identity is (11) with constant d. Differentiating $\rho V = u(c) + SV'$ with $c = \mathcal{R}^* + qS$ gives $\rho V' = u'(c)(\mathcal{R}_B + \mathcal{R}_q q' + q'S + qS') + S'V' + SV''$; at $S = 0$ the two terms in S' cancel by the first-order condition $V' = -qu'(c)$, leaving $\rho V' = u'(c)(\mathcal{R}_B + \mathcal{R}_q q')$. With $V' = -qu'(c)$, $\mathcal{R}_q = a(d)B$ and $\mathcal{R}_B = -(1-d)(\delta+\lambda) + qa(d) = -rq$ by the first identity, this reads $-\rho q = -rq + a(d)Bq'$, the second identity. The third is l'Hôpital's rule applied to $q' = (\xi(d)q - (1-d)(\lambda+\delta))/S$. □

Subsection A.5 Proofs for Section 3.4

Proof of Proposition 19 (sketch). The first two claims are as in Theorem 14. For (22), differentiate the HJB of state j in B at a point where $q_k \equiv 1$ and $d_k \equiv 0$ for every state, use $S_j = 0$ and the first-order conditions $-V_k' = u'(c_k)$. Inside a default-free region the HJB equation of a state $k \neq j$ with nonzero drift at B° is a regular first-order equation there, so V_k is C^1 at B°. The step that requires care is differentiating the HJB equation of state j itself, where V_j'' may exist only one-sidedly; the term $S_j V_j''$ vanishes at B° because $S_j(B^\circ) = 0$ and the one-sided second derivatives are finite, so the argument goes through with one-sided derivatives. □

Lemma 21 (the price is nonincreasing in the default path)**.** *Fix a z-path and two default paths with $d'_t \le d_t$ for all t. Then the Feynman–Kac values* (11) *satisfy $q(d') \ge q(d)$.*

Proof. For $\theta \in [0,1]$ let $d^\theta = d' + \theta(d - d')$ and $Q(\theta)$ the corresponding value. Differentiating under the integral and collecting the effect of d^θ_t on the payment at t and on the discounting of the stream after t (through $\xi'(d) = -\kappa(\delta+\lambda)$), $Q'(\theta) = \mathbb{E}\int_0^\infty e^{-\int_0^t \xi(d^\theta)}(\lambda+\delta)[\kappa q^\theta_t - 1](d_t - d'_t)\,dt \le 0$, since $\kappa q^\theta_t \le \kappa\bar{q} \le 1$. □

Proposition 22 (the best response preserves monotonicity in debt)**.** *Let q be nonincreasing in B and strictly positive, let the underwriting standard be slack so that the default rule is that of Theorem 3, and suppose the debt flow induced by the optimal policy under q is order preserving (two paths started at $B' < B$ with the same z-path satisfy $B'_t \le B_t$). Then the price generated by those policies through* (11) *is again nonincreasing in B.*

Proof. By Corollary 5 and the monotonicity of q, $d^*(B,z) = d^*(B,z,q(B,z))$ is nondecreasing in B; with order preservation, $d^*(B'_t, z_t) \le d^*(B_t, z_t)$ pathwise, and Lemma 21 gives the claim. □

Appendix B Numerical Implementation

Discrete equations. Debt lives on a grid B_i, $i = 0,\dots,n_B - 1$, with step ΔB, and productivity on z_j, $j = 0,\dots,n_z - 1$. One-sided differences are $D^+V_{ij} = (V_{i+1,j} - V_{ij})/\Delta B$ and $D^-V_{ij} = (V_{ij} - V_{i-1,j})/\Delta B$. At a node, for a given one-sided derivative V_B and price q, consumption is linear in the drift at fixed d, $c(S,d) = \mathcal{R}(d) + qS$, so the Hamiltonian $u(c(S,d)) + SV_B$ is concave in S with maximizer $S = (c^* - \mathcal{R}(d))/q$, $c^* = (u')^{-1}(-V_B/q)$, clamped to the admissible interval (from $c \ge c_{\min}$, $|S| \le \bar{S}$, the sign restriction of the candidate, and the underwriting standard $S \le -a(d)B$ where $q < \underline{q}$). Over d the maximized Hamiltonian is concave on $(0,1]$ and is maximized by golden-section search (28 iterations, a tolerance of about 10^{-6} in d), with the candidates $d = 0$, $d = 1$ and the closed form (10) also evaluated; the consumption floor is $c_{\min} = 0.05\,e^z$ and the drift cap $\bar{S} = 1$, and neither binds at any reported solution. Three candidates are formed at every node, forward (D^+V, $S \ge 0$), backward (D^-V, $S \le 0$) and zero drift; the one with the largest Hamiltonian is the policy. At B_0 the backward candidate is dropped and at B_{n_B-1} the forward one (state constraints). For a concave Hamiltonian this reduces to the standard upwind rule; with the fixed cost of default it is its monotone extension. The z-operator is discretized with upwinding of the exogenous drift and central second differences, so that every off-diagonal coefficient is nonnegative. With A the generator built from the policy, the discrete equilibrium is the root of

$$R_V \equiv \rho V - u(c) - AV = 0, \qquad R_q \equiv \xi(d)\,q - (1-d)(\lambda+\delta) - Aq = 0, \tag{27}$$

and all convergence statements refer to $\|R_V\|_\infty$ and $\|R_q\|_\infty$ at the current (V,q) with the policy recomputed from them, never to step-to-step changes.

Subsection B.1 Deterministic Case

All deterministic solutions reported in the paper are computed by the time stepping of Appendix B.2 with a single productivity state. The discrete system also admits an exact solution, which we use as a check and as a tool for the two-point problem of Section 3.3. Nodes with $S < 0$ depend only on the node below, nodes with $S > 0$ only on the node above, and a zero-drift node on nothing, so once the steady-state node is known the system is triangular and can be solved by marching outward. (i) *Candidates.* At each node, all prices q^* solving $q^* = (1 - d^*)(\lambda + \delta)/\xi(d^*)$ with d^* the maximizer of $u(c(0, d))$ at that price are located by a scan of $(0, 1]$ with bisection; $V^* = u(c^*)/\rho$. (ii) *Local consistency.* A candidate is kept if for some root of the node above, solved in backward mode, the forward candidate at the steady-state node does not beat zero drift, and symmetrically below. (iii) *Node solve.* Given the upwind neighbor, for each trial q the HJB node equation is monotone in V and solved by bisection, and the price equation becomes a scalar function of q whose roots are all located. (iv) *Branching march.* March up and down from the candidate, keeping every root that is consistent with the previous node's choice once the new value is known. (v) *Global check.* Verify at every node that the chosen candidate attains the maximum Hamiltonian given both neighbors, and evaluate (27). A solution is accepted if the residuals are below 10^{-8}, exactly one candidate survives with no violation, and the solution on $[0, \bar{B}]$ is unchanged when $\bar{B}$ is increased. At parameters with a small first default the march selects a unique candidate and reproduces the closed-form steady state and the cube-root constants to one or two percent. At the calibrated parameters, where the first default is half of the debt service, several locally consistent candidates survive step (ii) at neighboring nodes, the discrete form of the continuum of Lemma 20, and the march does not select among them; the time-stepped solution, which selects by the direction of time, agrees with the closed forms to a grid step (Table 6). The march also tests the two-point problem of Section 3.3 for candidate partial-default steady states: a candidate is a global equilibrium if the branch marched down from it reaches $B = 0$ without a violation.

Subsection B.2 Jump Process Case

The finite-state models are solved by joint implicit upwind time stepping with the policy explicit from (V^k, q^k),

$$\left[(\rho + 1/\Delta t)I - A^k\right]V^{k+1} = u^k + V^k/\Delta t, \qquad \left[(\xi(d^k) + 1/\Delta t)I - A^k\right]q^{k+1} = (1 - d^k)(\lambda + \delta) + q^k/\Delta t,$$

with $\Delta t = 0.01$, both solves sparse direct, initialized at the value of servicing the debt at the risk-free coupon forever, $u(e^z - \lambda B)/\rho$, and at the stationary price implied by the interior default policy at a price of one, until $\|R_V\|_\infty < 10^{-7}$ and $\|R_q\|_\infty < 10^{-8}$. The joint update matters: the alternating iteration (solve the HJB given q, then the price given the policies) has a loop gain of about thirty at the calibrated parameters and diverges. The upper end of the grid is placed where the drift is negative in every state, so that the truncation is exact; this is checked on the

solution. The pricing equation uses the same generator as the HJB equation, so its transport term is upwinded with the equilibrium drift: forward at B_0, where the drift is positive, and backward at B_{n_B-1}, where the state constraint makes it nonpositive. The stationary distribution solves $A^\top g = 0$ with the converged generator.

Subsection B.3 OU Diffusion Case

The diffusion case uses the same time stepping on the two-dimensional grid, with the z-operator discretized as above and Neumann conditions at $\pm\bar{z}$. In the calibration each parameter evaluation runs 3,000 implicit steps of $\Delta t = 0.015$ from a warm start at the previous point, interpolated onto the new z-grid, the first from the same initial guess as the finite-state models; the reported point is the calibrated parameters re-solved on the 75-node productivity grid from the calibration-grid solution, run until the moments are stable to the third digit. Moments are computed as described in Section 4.2: exactly from $A^\top g = 0$ for state functions, and by Euler–Maruyama simulation with bilinearly interpolated policies, 100 steps per year, annual aggregation of flows and reflecting boundaries for z, for path functionals. The impulse responses of Section 4.3 are expectations under the forward equation $\dot{g} = A^\top g$, started from the ergodic distribution pushed down in productivity by the shock (split linearly between neighboring nodes) and advanced with a Crank–Nicolson step of 0.01 years; a simulation with 20,000 paths gives the same responses to the third digit, with sampling noise visible only in the default intensity, which jumps at the threshold. The ergodic mass on nodes where the resolution ratio (28) below exceeds one half is reported with every solution.

A resolution artifact. The scheme obtains V_B by differencing, with an error of about $\frac{1}{2}|V_{BB}|\Delta B$, and the drift responds to that error with a gain of $1/(q^2|u''(c)|)$: consumption responds to the ratio V_B/q and the drift is the consumption gap divided by q again. Where the *resolution ratio*

$$\frac{\frac{1}{2}|V_{BB}|\,\Delta B}{q^2\,|u''(c)|\,|S|} \tag{28}$$

exceeds about one half, the scheme cannot determine the sign of the drift, and when it loses the sign it tends to produce a positive drift, which carries the state to the top of the grid. This looks like a second basin of attraction but moves with the grid: in the deterministic model the point beyond which the sign is unresolved moved from $B = 2.2$ to 2.6 to beyond 3 as the step was halved twice; in the calibration, a positive drift at average productivity above $B \approx 1.1$ on uniform grids disappeared on a grid refined where the price is small. The remedy is a grid whose spacing is proportional to q^2; we report the ergodic mass on unresolved nodes with every solution.

Appendix C Robustness

Grid and barrier, before the standard. Table 3 reports the point calibrated *without* the underwriting standard (the first of the two calibrations of Section 4.2; its parameters are in the notes), solved on four grids, to document the sensitivity that motivated the standard. The bulk is converged: the zero-drift loci at average and high productivity agree across grids to a grid step, and the mass in the debt bands $[0, 0.3)$, $[0.3, 0.5)$ and $[0.5, 0.7)$ is 0.48–0.52, 0.21–0.24 and 0.25 on the refined grids. The tail is not: the mass that escapes into the free-default explosion region (the top tenth of the grid in Table 3) is 0.9% and 5% on two refined grids ending at $B = 1$ and essentially zero on a refined grid ending at 1.5, because entry happens through a corridor at low productivity whose width is decided at the full-default cliff, where the drift sign chatters. This tail moves the mean partial default between 0.38 and 0.46, debt to output between 0.31 and 0.36, and the spread volatility between 0.022 and 0.058. On these grids the unconstrained sovereign in the free-default region issues at a rate of 0.3 to 0.5 of output per year, the price residual stalls between 0.3 and 0.6 on the cliff nodes, and the highest price at which it still issues in full default is 0.62 on the uniform grid, which the standard's floor must exceed (Table 4).

Table 3: The calibrated point without the underwriting standard on four grids

Grid	Nodes	$\bar{B}$	Loss	Mass top 10%	Mass full default	PD mean	Debt/ output
Uniform, $\Delta B = 0.005$	201	1.0	0.23	0.012	0.017	0.32	0.30
Non-uniform, $h_0 = 0.005$	977	1.0	0.46	0.009	0.023	0.40	0.33
Non-uniform, $h_0 = 0.0025$	1845	1.0	1.25	0.050	0.065	0.46	0.36
Non-uniform, $h_0 = 0.005$	2032	1.5	0.52	0.000	0.003	0.38	0.31

Notes: Parameters of the calibration without the standard: $\gamma_0 = 0.0354$, $\gamma_1 = 2.16$, $\gamma_2 = 0.126$, $z^* = 0.899$, $\rho = 0.0488$, $\kappa = 0.71$, $\delta = 0.124$, $\rho_z = 0.876$, $\sigma_\epsilon = 0.056$ (loss 0.228 on the uniform grid). Non-uniform grids have spacing $h(B) = \max\{h_0 \min_z q(B, z)^2, 2.5 \cdot 10^{-4}\}$. Mass top 10%: ergodic mass in the top tenth of the debt grid.

The level of the standard. Table 4 reports the same point solved with the standard at several levels. Floors of 0.4 and 0.5 do not remove the accumulation but relocate it to the debt level where the price falls through the floor, a trap inside the default region holding two to three percent of the mass; from 0.6 the top of the grid is empty, and at 0.7 no positive drift remains in full default anywhere.

Robustness of the calibrated point under the standard. Table 5 re-solves the calibrated point from a warm start on a finer debt grid, on the non-uniform grid of Section 3.2, and with the barrier at $B = 1.5$. With the standard in place the moments move by amounts comparable to the sampling noise of the simulated moments, the top of the grid carries no mass, and the loss is unchanged to the second digit.

Table 4: The calibrated point at several levels of the underwriting standard

$\underline{q}$	Spread at the floor (pp)	Mass in $\{S>0, d=1\}$	Max q in $\{S>0, d=1\}$	Mass $B \geq 0.9$	Mass full default
none	–	0.013	0.62	0.012	0.017
0.4	24.5	0.030	0.62	0.025	0.059
0.5	16.3	0.020	0.64	0.005	0.040
0.6	10.9	0.003	0.67	0.000	0.032
0.65	8.8	0.000	0.68	0.000	0.023
0.7	7.0	0	–	0.000	0.012

Notes: Uniform grid, $\bar{B} = 1$, parameters of the calibration without the standard. The spread at the floor is $(\delta + \lambda)(1/\underline{q} - 1)$, in percentage points.

Table 5: The calibrated point under the standard on other grids

Grid	Debt/y mean	Debt/y s.d.	Spread s.d.	PD freq.	PD mean	Loss	Mass top	Unresolved	Residuals
main grid: debt step 0.005 (201 nodes), $n_z = 75$, barrier at $B = 1$	0.236	0.166	0.027	0.378	0.422	0.237	0.0000	0.008	5e-03 / 1e-02
debt step 0.0025 (401 nodes), $n_z = 75$	0.236	0.166	0.027	0.377	0.423	0.239	0.0000	0.004	3e-03 / 1e-02
non-uniform debt grid (503 nodes, min step 1e-03), $n_z = 75$	0.236	0.166	0.027	0.377	0.423	0.238	0.0000	0.006	3e-03 / 1e-02
barrier at $B = 1.5$, $n_z = 75$	0.236	0.166	0.027	0.378	0.422	0.237	0.0000	0.008	6e-03 / 1e-02

Notes: Each row re-solves the calibrated point from a warm start on the grid indicated. Mass top: ergodic mass in the top tenth of the debt grid; unresolved: ergodic mass on nodes where the resolution ratio (28) exceeds one half; residuals: final HJB and pricing residuals.

Comparative statics. Table 6 re-solves the deterministic case across variations of the default cost, the recovery parameter, the discount rate and the maturity around the calibrated point, by the time stepping of Appendix B.2 on 200 nodes per unit of debt on $[0, 1.2]$, and compares the closed forms of Theorem 14 with the computed solution. In every case the first default node is the grid node that contains $B_{\text{pd}}(0, 1)$, the last node with positive drift is two nodes below it, and the computed default intensity at the first default node agrees with d_{first} to three digits. The steady state rises with γ_0, κ and γ_2 and falls with δ, as the closed form says and Corollary 5 predicts for the default cost and the recovery parameter, and it does not move with ρ: the discount rate changes only the price schedule beyond the steady state, and with it the points where the standard starts to bind and where default becomes complete.

Appendix D The Physics-Informed Neural Network Approach

Formulation. The natural neural network method for the diffusion case is the physics-informed neural network of Raissi, Perdikaris, and Karniadakis (2019), equivalently the deep Galerkin method of Sirignano and Spiliopoulos (2018): V and q are multilayer perceptrons of (B, z), derivatives are obtained by automatic differentiation, and the networks are trained to make the residuals of (23) and (24) small at random points of the state space. It is the continuous-time counterpart of the residual-based methods of Maliar, Maliar, and Winant (2021) and Azinovic, Gaegauf, and Scheidegger (2022), and it has been applied to continuous-time finance by Duarte,

Table 6: Comparative statics of the deterministic steady state

	Steady state		First default			Standard	Full default
Variation	$B_{\text{pd}}(0,1)$	computed	d_{first}	computed	$B_{\text{fd}}(0,1)$	binds from	from
Calibrated point	0.456	0.445	0.509	0.512	2.04	0.755	1.17
$\gamma_0 \times 0.8$	0.425	0.410	0.545	0.546	1.63	0.665	0.95
$\gamma_0 \times 1.2$	0.482	0.470	0.481	0.482	2.45	0.835	–
$\kappa = 0.45$	0.363	0.350	0.509	0.509	1.62	0.595	1.07
$\kappa = 0.65$	0.570	0.555	0.509	0.510	2.55	0.935	–
$\gamma_2 \times 0.5$	0.283	0.270	0.409	0.409	2.05	0.585	1.13
$\gamma_2 \times 1.5$	0.601	0.585	0.579	0.579	2.02	0.895	1.20
$\rho = 0.05$	0.456	0.445	0.509	0.512	2.04	0.785	1.19
$\rho = 0.07$	0.456	0.445	0.509	0.512	2.04	0.710	1.13
$\delta = 0.12$	0.597	0.585	0.509	0.511	2.67	0.915	–
$\delta = 0.22$	0.367	0.355	0.509	0.511	1.64	0.635	0.96

Notes: One parameter changed at a time from the calibrated point (Table 1), productivity at its mean, $\underline{q} = 0.7$. "Computed" steady state: the last grid node with positive drift; "computed" first default: the default intensity at the first node with $d > 0$. Residuals 10^{-7} and 10^{-8} in all cases but four ($\gamma_0 \times 1.2$, $\kappa = 0.45$, $\gamma_2 \times 0.5$, $\gamma_2 \times 1.5$), where the price residual stalled between 8×10^{-4} and 4×10^{-3} after 30,000 steps. "–": full default is not reached on the grid.

Duarte, and Silva (2024). For this model the residual route is attractive because the policies are available in closed form given (V_B, q) (Theorem 3), so the Hamiltonian is cheap at every sample point. In our prototype $V = \bar{V} + s_V N_V(x)$ and $q = \sigma(N_q(x))$ with N_V, N_q three-layer networks of width 64 with tanh activations, σ the logistic function so that $q \in (0,1)$; the residuals include an artificial viscosity εV_{BB}, annealed from $2 \cdot 10^{-3}$ to $2 \cdot 10^{-4}$, which plays the role of the numerical diffusion of the upwind scheme and without which the first-order transport term and the kinks of V make the loss ill-conditioned; 3,000 interior and 200 boundary points are resampled every step; the two networks are trained jointly on the sum of the residual losses, the price residual weighted by 25, with Adam.

Results. The experiment predates the calibration and the underwriting standard: it uses an earlier parameterization, a grid of 341×31 nodes on $[0, 1.7]$ and no floor, so its finite difference baseline (mean debt 0.36, instantaneous default probability 0.16) is not the model of Section 4; the conclusions concern the method. Four experiments on that case: a supervised fit of the networks to the finite difference solution, a residual run started from that fit, a residual run from random initialization, and a variant in which V_B and q_B in the residuals are one-sided differences of the networks chosen by the upwind rule of the finite difference scheme. Table 7 compares them to the finite difference solution on its grid.

What goes wrong. The networks have the capacity to represent the solution: the supervised fit reproduces V and q to $5 \cdot 10^{-3}$ and the moments to within a few percent. But every residual-trained network converges to a spurious solution of the stationary system, with residuals *smaller*

Table 7: Networks against the finite difference solution

	Supervised fit	Residual, from fit	Residual, scratch	Upwind residual
V rms error	0.005	0.46	0.46	0.45
q rms error	0.006	0.12	0.61	0.62
Drift sign agreement	0.93	0.47	0.43	0.43
Default region agreement	0.999	0.99	0.78	0.77
HJB residual rms	0.033	0.033	0.016	0.023
Price residual rms	0.017	0.012	0.010	0.012
Mean debt (FD: 0.36)	0.37	1.70	1.70	1.70
$P(d > 0)$ (FD: 0.16)	0.14	1.00	1.00	1.00

Notes: OU case on a 341×31 grid; errors are absolute, V ranges over $[-27, -25]$ and q over $[0.57, 1]$. The finite difference solution's own residuals are 10^{-7} (HJB) and 10^{-3} on a few nodes (price). Training times 2, 27, 77 and 70 minutes on three CPU threads; the finite difference solution took 76 minutes.

than those of the fitted network: started from the fit, it drifts to a price near one with borrowing everywhere; from scratch, to full default, a low price and all mass at the top of the grid. The stationary residual does not select the viscosity solution that the sovereign's problem defines, and neither the vanishing-viscosity continuation nor the upwind evaluation of the derivatives restores the selection that time stepping provides. Time stepping is a causal scheme, in which information travels along characteristics from the stationary points outward; a residual minimized over the whole state space has no such direction.

Promise for larger state spaces. The case for network methods is not precision in two dimensions, where the finite difference solution is tighter by orders of magnitude at comparable cost, but the growth of cost with the dimension of the state: a third state at the same resolution costs 30 to 300 times more on a tensor grid, and a fourth is out of reach, while the cost of a network sample grows linearly in the dimension and moments come from simulation. Three features of the model help in higher dimensions: closed-form policies given (V_B, q), bounded unknowns, and a stationary distribution concentrated on a low-dimensional region around the thresholds, where sampling can be targeted. Two hurt and will not go away: the kinks and cliffs, which require regularization in every controlled direction, and the need to select the right solution, which in our experiment only a causal scheme provides. A time-stepping network method, or a reinforcement learning formulation in which the networks are updated along simulated paths, is the route we see for the extensions discussed in the conclusion.